

UNIVERSITY OF MILANO-BICOCCA

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**Innovative Management of Service Requests:
From a Distributed Monolith to a Large-Scale
Microservices Architecture, with
RAG-Enhanced Recommendation System**

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“What I cannot create, I do not understand”
- Richard Feynman

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Introduction

The technological evolution of recent decades has profoundly transformed the business landscape, making IT solutions a crucial factor for the success of enterprises. In this context, Elmec Informatica S.p.A. stands out as a significant player in the Information Technology sector in Italy. Founded in 1971 and headquartered in Varese, Elmec combines extensive industry experience with a strong focus on innovation, offering services and solutions that cover a wide range of business needs, offering advanced IT solutions for medium and large enterprises.

The company manages four datacenters (including one Tier IV certified), provides Hybrid IT services, digital workspace and Device-as-a-Service solutions, and is a key partner for cybersecurity and regulatory compliance. With over 450 certified technicians, Elmec integrates cloud technologies (in collaboration with AWS and GAIA-X), promotes environmental sustainability, and stands out for its innovation through its Technological Campus and ongoing investments in professional training.

The primary context of this internship revolves around the redesign and enhancement of the existing *Service Request Portal (SRP)*, which is essential for managing client service requests efficiently.

A Service Requests is a formal request submitted by a customer to obtain specific services or actions within their IT infrastructure. These requests can encompass a wide range of activities, from creating new user accounts in the Active Directory system to managing user permissions, resetting passwords, and deploying software updates.

Various challenges have been identified within the current architecture, particularly concerning the dynamic forms used by customers for submitting the service requests, the management of customizable fields, and the integration with external data sources.

The main objectives of this internship are:

- **Analysis of the Existing System:** Conduct a comprehensive analysis of the current state of the SRP system, identifying weaknesses and inefficiencies that hinder its performance and user experience;
- **Architectural redesign:** Propose and implement a complete architectural redesign of the system;

- **Modular and Scalable Solution:** Design a modular and scalable solution that can accommodate various client-specific configurations;
- **Migrations:** Plan and execute a full migration of databases and other company services/platforms that use SRP.
- **Fine-tuned LLMs integration:** Explore opportunities for integrating specifically fine-tuned LLM systems into the SR workflow.

Chapter 1

Overview of the existing System

The internship project was not focused on creating an entirely new system from scratch, but rather on a deep redesign and restructuring of an existing system developed many years ago using legacy technologies and affected by several issues and critical limitations.

Before discussing the architectural, technological, and implementation choices that were made, it is necessary to carry out a thorough analysis of the current system, in order to understand its behavior, operational flows, technologies used, the main issues to be addressed, how to resolve them, how to prevent them from reoccurring in the future, and how to ensure that the new system is scalable and maintainable.

1.1 Service Requests Workflow

The lifecycle of a Service Request within the current system is orchestrated through a series of defined states and automated processes, involving interactions between various components and human operators. The flow is designed to manage SRs from initiation to completion, handling approvals, execution, and potential errors [27].

In Figure 1.1 [27], a simple and high-level flowchart is shown that illustrates the main flows and scenarios.

1.1.1 SR Initiation and Initial State Management

A Service Request can be initiated through two primary channels [27]:

- **Via MyElmec Portal:** Clients can submit an SR by completing a dedicated form from the MyElmec portal. Upon saving the form, a ticket is created in the HelpdeskAdvanced (HDA) system with a **QUALIFIED** status, and a corresponding Service Request is generated in SRP with a **NEW** status. These two entities are then linked via the HDA Ticket ID;
- **Via HDA Ticket by Elmec Operator:** Operators have the ability to associate a catalog SR directly with a ticket they are managing within the HDA interface. When the ticket is saved in a **DELEGATED** status, the SR

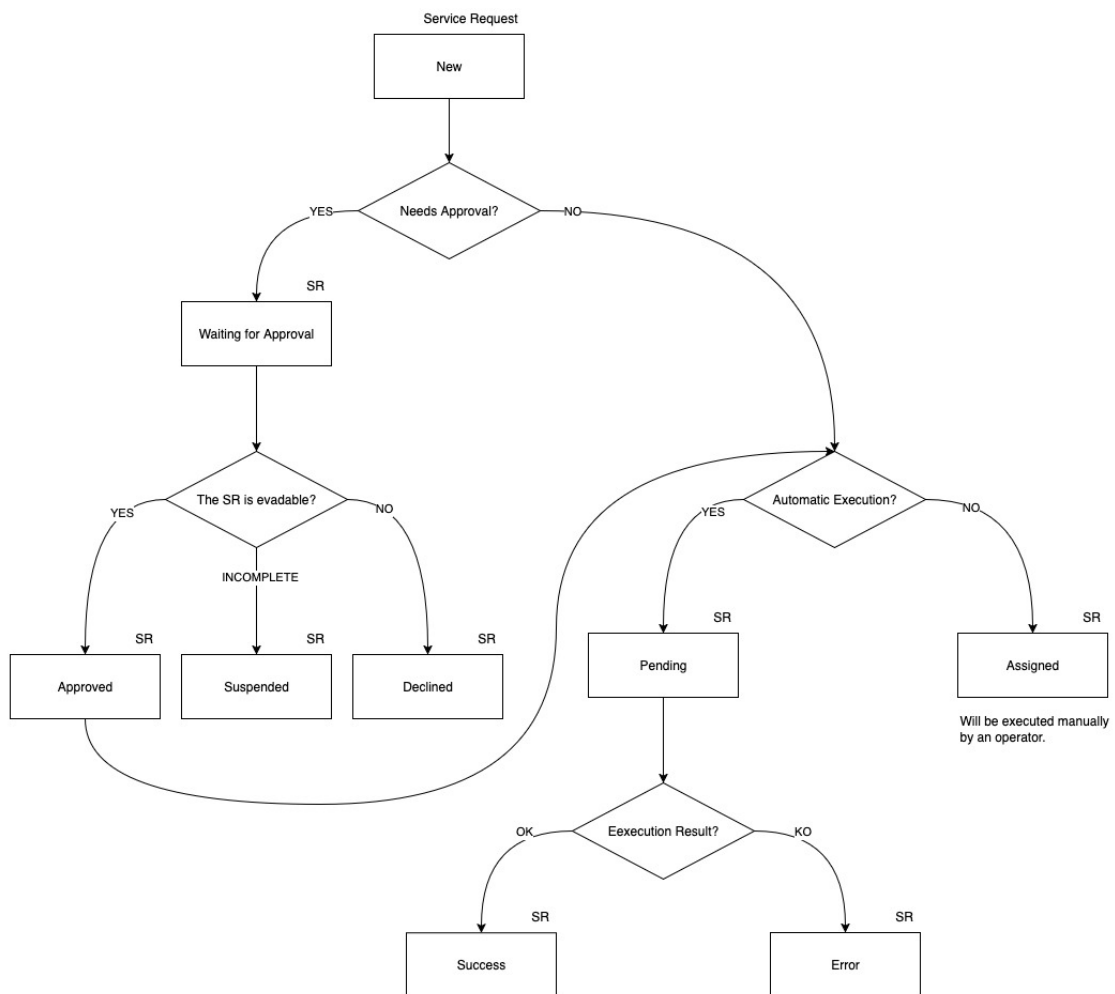


Figure 1.1: High-Level Service Request Lifecycle

is created in SRP with a **NEW** status, linked to the ticket, and immediately triggers the SR workflow within SRP.

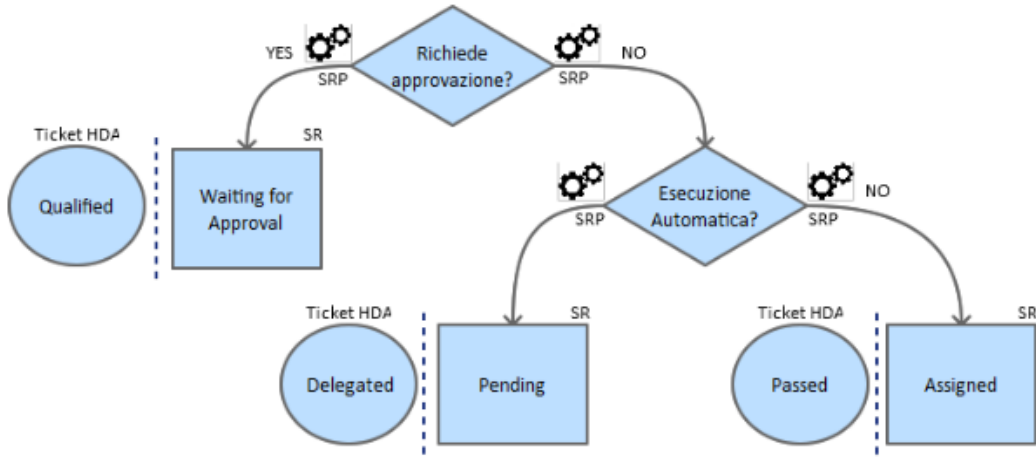


Figure 1.2: AS-IS New Service Request Workflow

As shown in Figure 1.2, following the Service Request creation, the system evaluates whether the Service Request requires approval by the operators. The logic for setting the initial SR and HDA ticket states is as follows [27]:

- **If approval is required:** the Service Request is set to **WAITING FOR APPROVAL** status, and the corresponding HDA ticket is set to **QUALIFIED**. A communication is added to the ticket, indicating that the SR needs validation from an operator before execution;
- **If no approval is required and it's linked to an automatism:** the Service Request enters a **PENDING** state, and the HDA ticket is set to **DELEGATED**;
- **If no approval is required and it's not linked to an automatism:** the Service Request is **ASSIGNED**, and the HDA ticket is set to **PASSED**. A communication is added to the ticket, noting that the SR requires manual processing by an operator.

1.1.2 Approval and Execution Workflows

As shown in Figure 1.3, for SRs requiring approval (**WAITING FOR APPROVAL** state with an HDA **QUALIFIED** ticket), an operator's intervention is necessary to validate the request. The operator has three possible actions [27]:

- **Cancel the SR:** the HDA ticket is set to **CLOSED ABORTED**, the SR status becomes **DECLINED**, and a notification email is sent to the requesting client;
- **Request client modification:** the HDA ticket is set to **WAITING USER**, the SR status becomes **SUSPENDED**, and a notification email is sent to

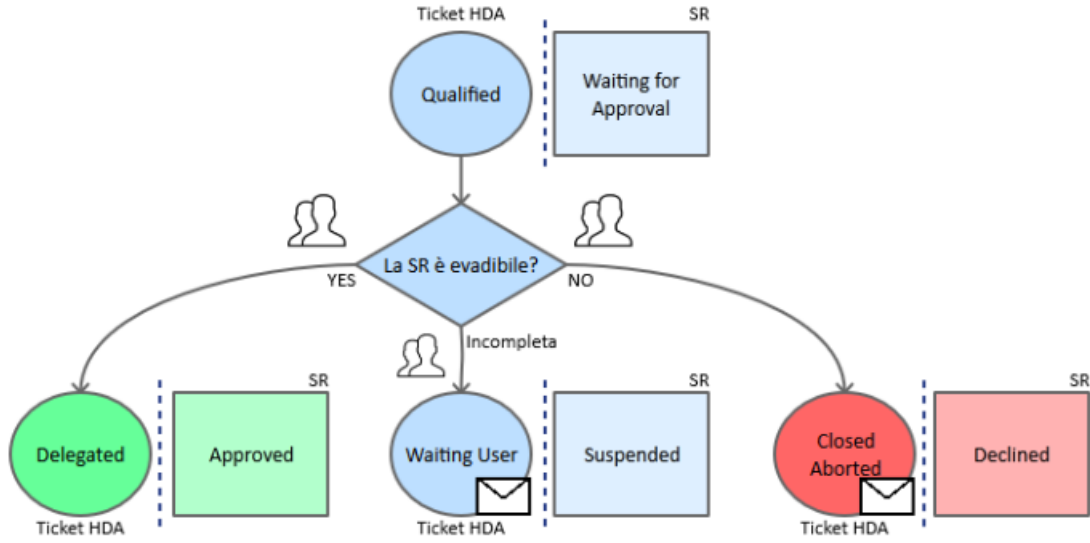


Figure 1.3: AS-IS Approval Service Request Workflow

the client. The workflow pauses until the client modifies the SR via Elmec.com;

- **Approve the SR:** the HDA ticket is set to DELEGATED, and the SR status becomes APPROVED.

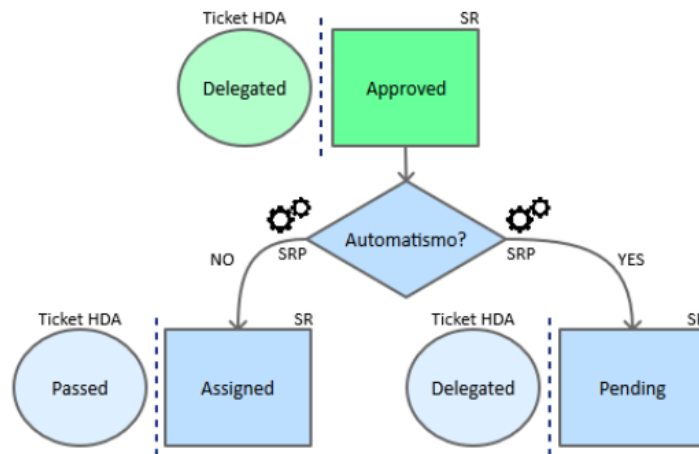


Figure 1.4: AS-IS Approved Service Request Workflow

As shown in Figure 1.4, once a SR is approved, SRP manages its subsequent flow based on whether it is linked to an automatism [27]:

- **If the SR is linked to an automatism:** the SR transitions to a PENDING state, while the HDA ticket remains DELEGATED. This implies it's awaiting automated execution;
- **If the SR is not linked to an automatism:** the SR is immediately set to ASSIGNED, and the HDA ticket is moved to PASSED. A note is added

to the ticket indicating that the SR requires manual processing.

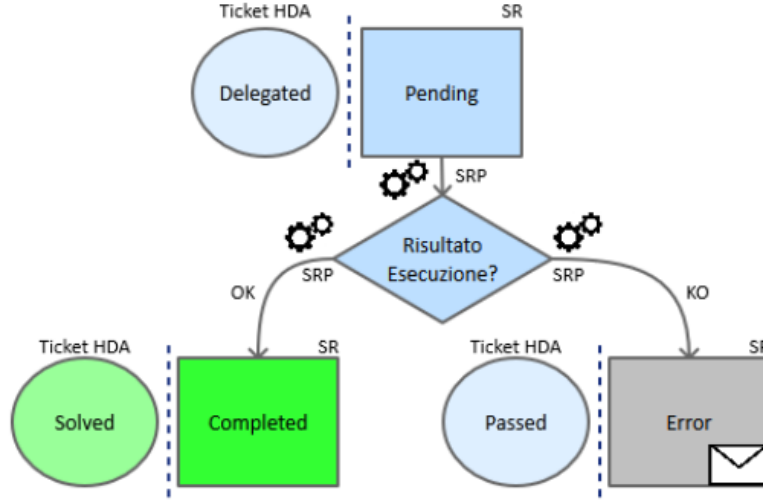


Figure 1.5: AS-IS Pending Service Request Workflow

As shown in Figure 1.5, an SR in PENDING state is awaiting the execution of its associated automatism. The outcome of this automated execution determines the next state [35]:

- **Successful execution:** the SR is set to COMPLETED, and the corresponding HDA ticket is set to SOLVED;
- **Unsuccessful execution:** the SR transitions to an ERROR state, and the HDA ticket is set to PASSED. Communication regarding the error is added to the ticket for visibility within HDA.

The execution of Service Requests themselves can involve the SRP Agent on the client's Domain Controller for operations requiring client-side interaction (e.g., Active Directory or Exchange environment changes). The SRP Agent periodically checks ELMEC-SRP01 for pending SRs. If found, it retrieves the SR details, loads the relevant PowerShell script, and executes it. The agent then informs ELMEC-SRP01 that the SR is running [35]. Monitoring is performed by a scheduled READLOG task on the client side, which checks the PowerShell script's log file every minute. A "KEY" in the log indicates completion, prompting the SRP Agent to inform ELMEC-SRP01 to set the SR to COMPLETED, triggering an HDA status update. An "ERROR" key indicates failure, leading to the log being sent to ELMEC-SRP01 and an error signal. A timeout mechanism also signals an error if completion is not detected within a specified duration [35].

A constraint that must be respected is that the high-level flow (state transitions and mapping between SR states and HDA ticket states) must not be modified, as it represents an established and approved process that also encompasses organizational and accountability aspects among various Elmec departments

[28]: those who configure and manage the catalog and related Service Requests (Managed Services (MxS) team), those who develop and maintain the product (Innovation team), and those responsible for the ticket management system (HDA team). The technological and implementation aspects, on the other hand, can and should be redesigned and improved [28], as will be explored in the following chapters.

1.2 Current System Architecture

The *SRP* system is fundamentally divided into two primary segments:

- Company Network;
- Client Network.

These segments interact and communicate exclusively through a *DMI Console* [29], which acts as a unidirectional bridge, transferring requests from the client network to the internal company network [29] [14].

The overall architecture is depicted in the diagram in figure 1.6 [29].

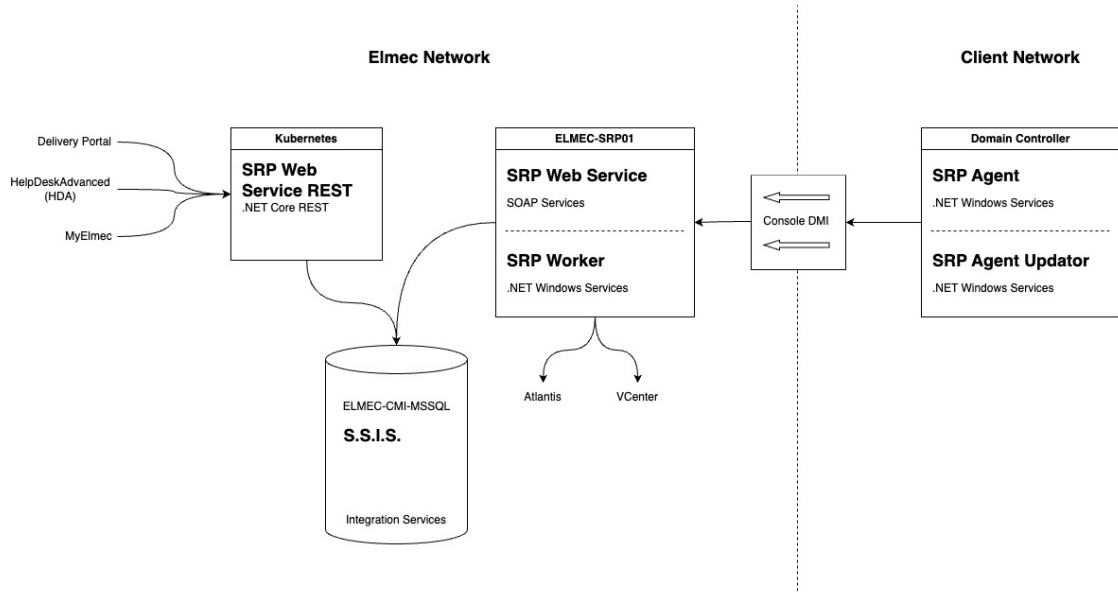


Figure 1.6: AS-IS System Architecture

The system's operational flow involves several key components across both networks, orchestrating the processing of service requests.

1.2.1 Main Functionalities and Components

The current *SRP* system relies on a suite of interconnected components, each serving a specific role in the lifecycle of a service request.

Elmec's Network Components:

- **SRP Web Service REST** [34]: This component, deployed on a Kubernetes cluster, exposes .NET Core RESTful services [29]. It serves as the primary interface for external systems such as:
 - **Delivery Portal**: An asset management system used to manage Service Request catalog configuration and Elmec’s internal and client servers;
 - **HDA (HelpDeskAdvanced)**: A ticket management system. Each service request is intrinsically linked to an HDA ticket via an ID, and HDA is responsible for initiating and associating service requests with existing tickets.
 - **MyElmec**: A client-facing portal enabling customers to create, manage, and monitor the various services provided by Elmec (e.g., servers, management software, cloud solutions).

The *SRP Web Service REST* communicates with the *ELMEC-CMI-MSSQL* server.

- **ELMEC-CMI-MSSQL**: This server hosts the central SQL Server Database for the SRP system. It is the persistent storage layer for all service request data and related information [29].
- **S.S.I.S. (SQL Server Integration Services)**: Running on the ELMEC-CMI-MSSQL server, SSIS packages are crucial for data transformation and import/export operations. They facilitate the transfer of data, such as master data (e.g., Active Directory users, Organizational Units, groups) sent by SRP Agents, into the database. Each SSIS package is designed to transform the received information for database insertion or vice versa [29].
- **ELMEC-SRP01**: This Windows Server functions as the core engine for service requests within the Elmec network. It houses all the PowerShell scripts associated with individual service requests, which are maintained by the Platform team [29]. Key components residing on ELMEC-SRP01 include:
 - **SRP WebServiceSoap**: This component provides SOAP services primarily for communication with the SRP Agent located on the client side. It interacts with the database to retrieve necessary information before exposing it to the agents for service request execution [33].
 - **SRP InternalWorker**: A Windows service that operates similarly to the SRP Agent but executes entirely within the Elmec network. This is utilized for service requests that do not require interaction with the client’s network or Active Directory (e.g., "no patch" service

requests). The InternalWorker queries the database every 10 seconds for "internal worker" type jobs (like removing a server from a patching session, or resetting a managed Virtual Machine), connects to the SOAP service, retrieves the Service Request ID and its fields, and initiates job execution. It also monitors logs every 30 seconds to determine the job's status [32].

The only component in the client's network is called **Domain Controller**, which represents the client's Active Directory domain controller, where the communication tools with Elmec are installed. It hosts:

- **SRP Agent (.NET Windows Service)**: Typically, one agent is configured per client domain, specifically for Active Directory (AD) and Exchange-related service requests. This agent performs two main functions [35]:
 - Every 8 hours, it reads master data (AD users, OUs, groups) from the domain controller and sends it to ELMEC-SRP01 for storage via SSIS.
 - Every 5 minutes, it queries ELMEC-SRP01 for pending Service Requests. Upon finding any, it requests detailed information, loads the corresponding PowerShell script, and executes it. After initiating the script, the agent informs ELMEC-SRP01 that the SR is "running" and proceeds to the next SR.
- **SRP Agent Updator (.NET Windows Service)**: This service continuously monitors the SRP Agent folder for a file with a .new extension. If detected, it stops the SRP Agent, updates it, and restarts it

1.2.2 Interactions and Communication between Customers and Company networks

The Communication between the company network and the clients' networks is a critical aspect of the SRP system's operations and is handled through the **DMI Console**.

The DMI console serves as a key infrastructural component, primarily acting as a communication bridge between the customer's network and Elmec's internal network. Specifically, for the SRP system, the DMI console is responsible for transferring service requests (SR) from the customer's network to Elmec's internal network, although it does not handle traffic in the opposite direction [14].

Operation of the DMI Console

The connectivity of the DMI appliances to Elmec is established through multiple OpenVPN connections, initiated by the appliance towards the Brunello 4 and Mendrisio Data Centers (for Swiss customers). The DMI consoles are not directly exposed to the Internet and are accessible only by authorized personnel. Hardware requirements for a DMI node include 4 GB of RAM, 4 cores, and 50 GB of disk space [14].

Following recent developments in the DMI, the new consoles use K3S for container management. K3S is a lightweight, certified Kubernetes distribution designed for edge, IoT, and CI/CD environments. It is characterized by being a single binary that reduces external dependencies and uses SQLite as the default database, greatly simplifying installation and management. It already includes essential components such as *containerd* [18].

All consoles are centrally managed via Rancher, and the deployment of the application stack is delegated to ArgoCD, which ensures that container versions are constantly updated. For security reasons, the console in the customer network by default exposes only SSH, while services such as SRP are selectively enabled and only when necessary. All application data present on the consoles is stored on a LUKS (Linux Unified Key Setup) encrypted volume¹, which can only be unlocked if the VPN tunnel is established. The operating system update is performed through a standard Elmec system, called PMD (Patching Management Dashboard) [14].

1.3 Problems and core issues of the current system

The analysis of the architecture and workflow of the current SRP system has revealed a number of critical issues that compromise its scalability, maintainability, and reliability. These problems are deeply rooted in the technological and design choices made years ago and are evident across several key areas of the system, from catalog management to script execution and the import of master data from Active Directory.

¹A platform-independent hard disk encryption method that can be used to encrypt disks across a wide variety of tools. This not only ensures full compatibility and interoperability between different software but also guarantees that password management occurs in a secure and documented manner [9].

1.3.1 Architectural and Technological Challenges

The SRP system is characterized by a fragmented architectural structure and complex interdependencies between its main components, which leads to several critical issues.

Monolithic Frontend Component

The user interface, responsible for navigating the Service Request catalog and completing the associated forms, is an excessively large and complex Vue.js frontend component (nearly 11,000 lines). This architecture makes modifications, extensions, and client-specific customizations exceptionally challenging [20]. The logic for managing dynamic fields, their interdependencies, and the rules for compilation and validation are all handled at the frontend level through a long series of hard-coded conditional statements, such as the following:

```
1 <div v-if="field.desc == 'Domain'">
2   ...
3 </div>
```

This, in addition to mixing business logic with presentation, represents a significant challenge for catalog configuration, as there is no centralized way to manage the fields and their relationships. Every change, therefore, requires a direct intervention on the code, making the system inflexible and difficult to maintain, especially in a context where the operator modifying the catalog is not a programmer and does not know how the frontend code is structured. For instance, if an operator wanted to add a new "Domain" field to a Service Request, this field would need to have the exact description "Domain" and type "domain"; otherwise, the Vue.js component would not render it correctly.

These dynamics make the system fragile and susceptible to frequent errors that waste time and resources for both clients and operators [20].

Overly Generic Backend REST Service

The only information about the fields returned by the backend REST service (SRPWebServiceREST), besides the field name and its mandatoriness, is its *type*. It is therefore easy to imagine how this information, which should only be an indication of the component that needs to be rendered (e.g., textbox, select, radio button, etc.), can become an identifier for something much broader.

For example, a field of type UPN is a composite field derived from the value of another field, 'Display Name', concatenated with an '@' symbol, and suffixed with the value from a select field whose options are the client's domains, provided by an external REST service (e.g., "n.guarini@elmec.it", where "n.guarini" is the value of the "Display Name" field, and "elmec.it" is the

value of the "Domain" select field).

Despite this complexity, the REST service returns only a JSON document of this type:

```
1 {  
2     "idField": 1603,  
3     "descField": "UPN",  
4     "type": "upn",  
5     "mandatory": true  
6 }
```

thus delegating to the frontend the responsibility of interpreting the type and rendering the field correctly.

Dependency on the SRP Agent

The import of data from clients' Active Directories and the actual execution of Service Requests are handled by an agent (`SRPWindowsService`) installed directly on the clients' Domain Controllers. This approach presents several challenges:

- **Security and Permissions:** The agent often requires credentials with elevated privileges (e.g., `domain-admin`), which clients are increasingly reluctant to grant.
- **Reliability and Control:** The configurations on the agent and the Domain Controller are complex and difficult to manage centrally, which can cause execution errors and data synchronization issues.
- **Scalability:** The need to install, configure, and maintain an agent for each client domain represents a significant operational effort. Alternatives such as more modern provisioning systems will be evaluated.
- **Obsolete Communication Patterns:** The unidirectional communication through the *DMI Console* creates a bottleneck and complicates the management of asynchronous operations. Communication between components in the client's network and the company network is based on polling (e.g., the SRP Agent queries the server every 5 minutes), an approach that introduces latency and inefficiencies.

1.3.2 Code and Data Management Issues

Code and data persistence management is another critical aspect of the system, characterized by a lack of versioning and modern debugging tools.

Unversioned and Undebuggable Stored Procedures

The intensive use of Stored Procedures² in the MSSQL database represents one of the major weaknesses. These procedures contain fundamental business logic and are:

- **Unversioned:** There is no centralized version control system for Stored Procedures. Changes are made directly on the database, making it impossible to track changes, revert to previous versions, or test modifications in separate environments.
- **Difficult to Debug:** Debugging Stored Procedures is a complex and cumbersome process, often requiring direct access and specific SQL server tools, with the risk of affecting the production environment.

Inaccessible and Unversioned SSIS Jobs

SSIS packages³ are essential for data import and export operations (e.g., AD master data). However, these jobs reside physically on the ELMEC-CMI-MSSQL server, and the only way to access and modify them is via a remote desktop. This approach prevents versioning and collaboration, limits accessibility by making maintenance and debugging an onerous and risky activity, and creates a *single-point-of-failure* and a strong coupling between the import/export business logic and the infrastructure.

1.3.3 Database Schema and Data Integrity Issues

The analysis of the database schema, represented in figures 1.7, has revealed a number of significant issues that affect the quality and integrity of the data managed by the SRP system.

Lack of Data Integrity and Consistency

One of the most significant problems concerns the lack of explicit referential integrity constraints. In the schema in figure 1.7, although the relationships between tables have been inferred manually, they are not defined at the database level through the use of foreign keys. This absence prevents the

²Stored procedures are SQL code routines, or sets of instructions, that are pre-compiled and stored within a relational database. They act as logical units of execution, encapsulating business logic directly at the database level [37].

³SQL Server Integration Services (SSIS) is a platform for building data integration and workflow solutions. It is a key component of Microsoft SQL Server, primarily used for Extract, Transform, Load (ETL) operations, enabling the extraction of data from various sources, its transformation, and its loading into different destinations [31].

database from guaranteeing data integrity [7]: for example, a row in a child table could refer to a non-existent row in the parent table [45]. Such a scenario, known as a "dangling reference"⁴, can generate anomalies and inconsistencies in the system, making the data unreliable and debugging extremely complex [7].

Data Redundancy

Another issue is the high data redundancy and the large number of duplicate attributes across different tables [35]. This design, which deviates from the principles of normalization [2], leads to storing the same information in multiple locations, which not only increases the data volume but also introduces a high risk of inconsistencies [2].

Unclear and Inconsistent Naming Conventions

The naming convention for attributes and tables is inconsistent and unclear. The schema presents a mixture of conventions (e.g., `IDGroup` vs. `GroupID`, English names mixed with specific terms) and typos (e.g., `TBParamenters`), making it difficult to immediately understand the function of each field. In many cases, the naming is not explicit enough to describe the role of an attribute, requiring an in-depth knowledge of the system to correctly interpret the relationships and business logic. This lack of a unique and standardized vocabulary hinders collaboration among developers, complicates code maintenance, and increases the likelihood of errors.

1.3.4 Maintenance and Operational Challenges

According to information discussed in an internal meeting [28], the daily management of the system is made complex by several inefficiencies and obsolete design choices.

The current system design does not allow for the effective reuse of Service Requests and catalog configurations. Consequently, every modification or addition requires the creation of a new Service Request, even for small variations, leading to a inefficient and difficult-to-manage catalog, consisting of numerous duplicate SRs and fields. This, while functional, is not a scalable approach and leads to significant inefficiencies in catalog management itself. For example, if a Service Request needs to be modified for all clients who use

⁴The "dangling pointer" is a problem typically addressed in the context of low-level programming, where a pointer refers to a memory area that is no longer valid because it has been freed or because the pointer is used outside the context of the variable's existence to which it refers [16], but the same dynamic can also be applied in the context of databases, where the "pointers" are represented by foreign keys.

it, the operator will have to modify not only the SR in question but also all the duplicate SRs for each client, with the risk of forgetting a modification or making mistakes. This approach, although probably conceived to prevent error propagation, makes the catalog difficult to navigate and maintain.

The new architecture will need to address these challenges by introducing a more robust, versioned system based on modern software design principles and patterns.

Chapter 2

Architectural Redesign

Given the numerous critical issues that emerged from the analysis of the current system, it is clear that a thorough redesign is necessary. This redesign must be able to perform the system's current functions more efficiently and in an optimized manner, while also satisfying a series of requirements that did not exist when the system was developed several years ago, and which the current system cannot manage because it was not designed to do so.

2.1 New System Requirements

The system can be divided into three main components: **Catalog Structure and Management**; **Service Request Execution**, and **Master Data Import**. The requirements for each of these areas will therefore be analyzed.

2.1.1 Catalog Structure and Management

This is the most substantial component of the three, dealing with everything concerning the definition, maintenance, and configuration of the actions executable by each customer. This therefore also includes the management of form fields, customer-specific customizations, scripts, their parameters, the target machine address retrieval logic, and so on.

Before proceeding with the analysis of the requirements, a clarification on the terminology to be used is necessary: an '*Action*' is understood as the formal definition of an operation that a client can execute on its infrastructure; a '*Service Request*' (SR), on the other hand, is understood as the effective execution of an Action. If a parallel with OOP is to be drawn, the Action can be seen as a class, and the Service Request as an object, an instance of the aforementioned class.

Actions

The Action entity can be seen as the core of the catalog component, and it is composed of the following main elements:

- General metadata, such as the name, an extended description, tags, a type, a group, a category, and other information related to administrative

and billing aspects;

- The fields of the corresponding form that must be filled by the client for its execution, including the relative logic tied to types, data validation, dependencies between various fields (e.g., fields that, once completed, "enable" other fields), autofill (e.g., fields whose values are autofilled with values from other fields), sensitive values, and so on;
- The script that will be executed in the customer's infrastructure to perform the effective required operation;
- Script parameters, which may be field values, but also additional parameters necessary for execution (e.g., AWS credentials, 365 credentials, etc...);
- The target machine on which the script must be executed.

Some of these dynamics are already implemented in the current system (with all the critical issues of the case, as already discussed in the previous chapter), while others are not. The crucial point, however, is that all these configuration aspects must be customizable for specific customers and for specific actions [28].

Given that the users of this system are very large clients (Fendi, Valentino, Lavazza, to name a few), it is easy to imagine that each of them has highly specific needs, which make direct reuse of Actions across multiple clients impossible. At the same time, however, the creation of infinite duplicated Actions that share almost everything, with only small differences (as happens in the current system [28]), must be avoided.

It is therefore necessary to implement a system that provides for the possibility of defining a sort of hierarchical structure, where a "parent" action is configured, which can be "inherited" by several child actions that will specify only the modifications, all without affecting the default action.

Fields

Together with actions, fields are central components on which a large part of the system is based. As already mentioned, fields are entities that can be extremely diverse, such as text fields, selects, multiple-choice selects, radio buttons, etc. In the current system, there are 86 different field types. This number is so high because, due to how the current system is built, the "type" information is the only way the backend communicates to the frontend what type of field is to be rendered [34]. In fact, there are many different types that share the same logic, perhaps simply using two different services to retrieve the data. The effective field typologies are therefore much fewer, and it is crucial that they are identified and generalized as much as possible.

The general idea is that the logic for fields must be moved to the backend, which must return all the necessary information for the frontend to render the form: from field types, to validation regexes, to error messages, to any external services to be contacted to retrieve the options list, or to validate the data, and so on [28].

All this must obviously be compatible with the customization requirements specified in the previous section.

Regarding customizations, a typical example scenario could be the following: a customer uses an action, but for some reason requires two additional fields, which will then be needed by the script that will in turn require two additional parameters. Furthermore, for some fields, the client in question uses different validation rules and specific autofill templates. These customizations must be possible without modifying the default action and fields.

Regarding "particular" field types, the system must provide for [28]:

- Select fields whose options are customer-specific (e.g., "Domain" field, which is a select with all the customer's domains);
- Select fields whose options are global for the entire system (e.g., "Country" field, which provides a list of states that are the same for all customers);
- Select fields whose options are customer-specific and to obtain them an external service must be contacted, with the possible option to search (e.g., "Server" field, which is a select with all the customer's servers, returned by the Atlantis CMDB service);
- Autofilled text fields, meaning their values are obtained through a templating system (Jinja) that uses the values of other fields (e.g., "Logon Name" field, populated by concatenating the user's first and last name separated by, for example, a dot);
- Text fields whose validation occurs through regex (e.g., "Date" field, which must comply with the standard format);
- Text fields whose validation occurs through an external service (e.g., "Logon Name" field, which requires contacting the relevant Active Directory service for validation that a user with the same logon name does not already exist);
- Fields with sensitive values, whose values will therefore need to be saved encrypted, using some symmetric encryption algorithm (e.g., "Password" field);
- "Mixed" fields, composed of multiple "sub-fields" with different characteristics (e.g., "UPN" field, which is composed of the first letter of the

first name, the last name, an at sign (@), and the value of a select whose options are obtained from an external API call);

It is important that the system is designed in a scalable way, so that if another field typology were to be added in the future, it can be implemented with the minimum possible effort.

Configuration and Execution Consistency

A system for monitoring the consistency of the catalog configurations [28] will be necessary, in order to prevent the creation of invalid configurations, which would then lead to errors during SR execution.

For example, if a **select** type field is specified for an action, whose options are obtained from customer-specific parameters, but those parameters have not been defined for that customer, the system must report this inconsistency through some alerting system (e.g., email, ticket, monitoring dashboard). Alternatively, if an action is configured to be executed on a certain type of machine but the field necessary to select it is not among the action's fields, the system must report this inconsistency.

Similarly, a system for monitoring the consistency of SR executions must be implemented, in order to be able to report any errors that may have occurred during execution, such as an SR stuck in an "intermediate" state (e.g., **NEW**) for too long, or an SR that started execution (i.e., in **PENDING**) but never reached completion (i.e., in **COMPLETED** or **ERROR**) within a certain time limit (e.g., 2 hours).

Additional script parameters

Another topic that must be managed is that of script parameters. In the old system, there was granular control over every single parameter, which was linked to an action's script and possibly to a field in the corresponding Service Request. In practice, all the action's fields with their respective values were passed as parameters to each script, plus some additional parameters not linked to any field (e.g., Exchange Server, AD Sync Server, various credentials). This dynamic of additional parameters occurred through the definition of parameters with the foreign key on the field set to **NULL**, which were then intercepted by the stored procedure relating to the generation of the parameter string which, with hard-coded **if** statements (e.g., **IF @ParamName = 'ServerExc' BEGIN ...**), were valued accordingly, retrieving the information with custom and non-standardized logic.

In the new system, it will be necessary to standardize these dynamics, finding a way to define the additional parameters that will be needed, beyond those of the fields, during action configuration, and to generalize the implementation logic as much as possible, so that if new "groups" of additional parameters

(e.g., AWS credentials, O365 credentials, etc.) were to be added, it can be done in the most efficient way possible.

Target Machine Configuration

In the old system, Service Requests were executed exclusively on the customers' Domain Controllers. Specifically, each DC ran an agent, which periodically checked via *polling*¹ if there were any Service Requests in the **PENDING** state with a matching **domain** field.

This part of the architecture will also need to be redesigned, as it is inefficient and highly limiting by design [41]. Therefore, in addition to redesigning these dynamics, it will also be necessary to provide the option to choose, during the configuration of an Action, to execute the related SRs on other types of machines, such as internal Elmec workers, or other types of machines on the customer's network other than the Domain Controller (e.g., Exchange Server). It must also be possible to choose to execute SRs through the old flow, in order to continue supporting both systems, at least initially [28].

2.1.2 Service Request Execution

In the current system, the execution of SRs occurs according to this sequence of main events:

1. The frontend sends the request to create a new SR to the REST service (**SRPWebServiceREST**), with all the necessary data (completed fields, customer, action, etc.) [34];
2. The SR and the relevant field values are created in the database, setting the status to **NEW** [34];
3. An SSIS job, every minute, checks if there are SRs in the **NEW** state, and if any are found, they are processed, moving them to subsequent states (e.g., **WAITING FOR APPROVAL**, **PENDING**, **ASSIGNED**, etc.) depending on the action and customer configuration [29];
4. When the SR is in the **PENDING** state, it means it is ready to be executed by the agent on the customer's DC;
5. Every 10 seconds, the agent on the customer's DC checks if there are SRs in the **PENDING** state for its domain (through calls to the SOAP service (**SRPWebService**)) [35];

¹*Polling* is a communication technique where a system or device periodically and actively samples the status of an external resource or device to check for readiness or new data.

6. If any are found, for each one, the corresponding script is downloaded, executed, and the SR status is updated to **COMPLETED** or **ERROR** depending on the execution outcome (through the SOAP service) [35];

It has already been discussed in the previous chapter how this flow has numerous criticalities, which make it inefficient and not very scalable. It will therefore be necessary to completely redesign it, taking into consideration the possibility of executing SRs on machines other than DCs, and the possibility of executing them also on machines internal to Elmec, and even virtual machines [28].

Elmec already has an internal *provisioning* system, called *Provision Prime*, which is responsible for executing Ansible playbooks on machines (internal, external, and also virtual) [24], and which could be leveraged for this purpose. It will therefore be necessary to analyze this system, and understand if and how it can be integrated into the new SR execution flow, in order to avoid the polling of agents on the DCs, and to have a generally more reliable and efficient system [28].

A constraint that must be respected, however, is that the action scripts are all in PowerShell, and it is not possible to modify them. This is because they are very numerous, have been developed over the years, have been tested and validated for every single client, and are maintained by different Elmec departments. Therefore, even if the SR execution flow is redesigned, the scripts must be executed as they are. This obviously does not apply to new scripts that will be developed in the future, which can be designed leveraging the new technologies [28].

2.1.3 Master Data Import

Another important component of the system is the one responsible for importing customer master data, necessary for populating some action fields (e.g., UPN domains, user accounts, groups, Organizational Units (OU)², databases, etc.). The main difficulty lies in the fact that this information physically resides on the customers' machines (large enterprise clients, with thousands of user accounts and groups, are being discussed), and therefore their retrieval in real-time is not immediate.

In the current system, this import occurs in a manner very similar to the SR scenario, but in reverse. It happens through an agent running on the customers' DCs, which periodically (hourly) (downloads and) executes a series

²In Windows Server, an Organizational Unit (OU) is a container within Active Directory (AD) used to logically group objects like user accounts, computer accounts, and security groups, similar to a folder in a file system [1].

of PowerShell scripts to retrieve the necessary information, and deposits them in a specific system folder (D:\SRPortal\Anag\) [35]. These files are then consumed by an SSIS job (one for each type of master data) that imports them into the database. The REST services then expose APIs to retrieve this information, which are used by the frontend to populate the action fields [34].

In the new system, it will be necessary to redesign this flow as well, in order to eliminate the dependence on agents on customer DCs. Even in this case, however, the PowerShell scripts cannot be modified [28], and must be executed as they are now. It will therefore be necessary to analyze how to integrate the execution of these scripts into the new flow, perhaps also leveraging Elmec's internal *provisioning* system (*Provision Prime* [24]) in this case, in order to obtain a more traceable, reliable, and efficient system [28].

2.2 New System Architecture

After gathering and defining the requirements for the new *ServiceRequestManager* (SRM) system through various meetings with stakeholders and analysis of the current system (SRP), it has been possible to design a new architecture for the system that allows for efficient and scalable satisfaction of these requirements.

The proposed new architecture for the SRM system, shown in Figure 2.1, is based on a **microservices architecture**, which allows for greater modularity, scalability, and ease of maintenance compared to the distributed-monolithic architecture of the current SRP system [11].

2.2.1 Main Components

The SRM system architecture is composed of the following main components.

Frontend

Two separate web frontends will be developed for end-users and system administrators. These frontends will communicate with the REST services via HTTP/HTTPS APIs.

Due to the business logic being moved to the REST services, the frontends will be lighter and easier to maintain, allowing for greater flexibility in implementing new functionalities and user interface improvements [20]. This also allows for the future integration of a mobile application, which can use the same APIs.

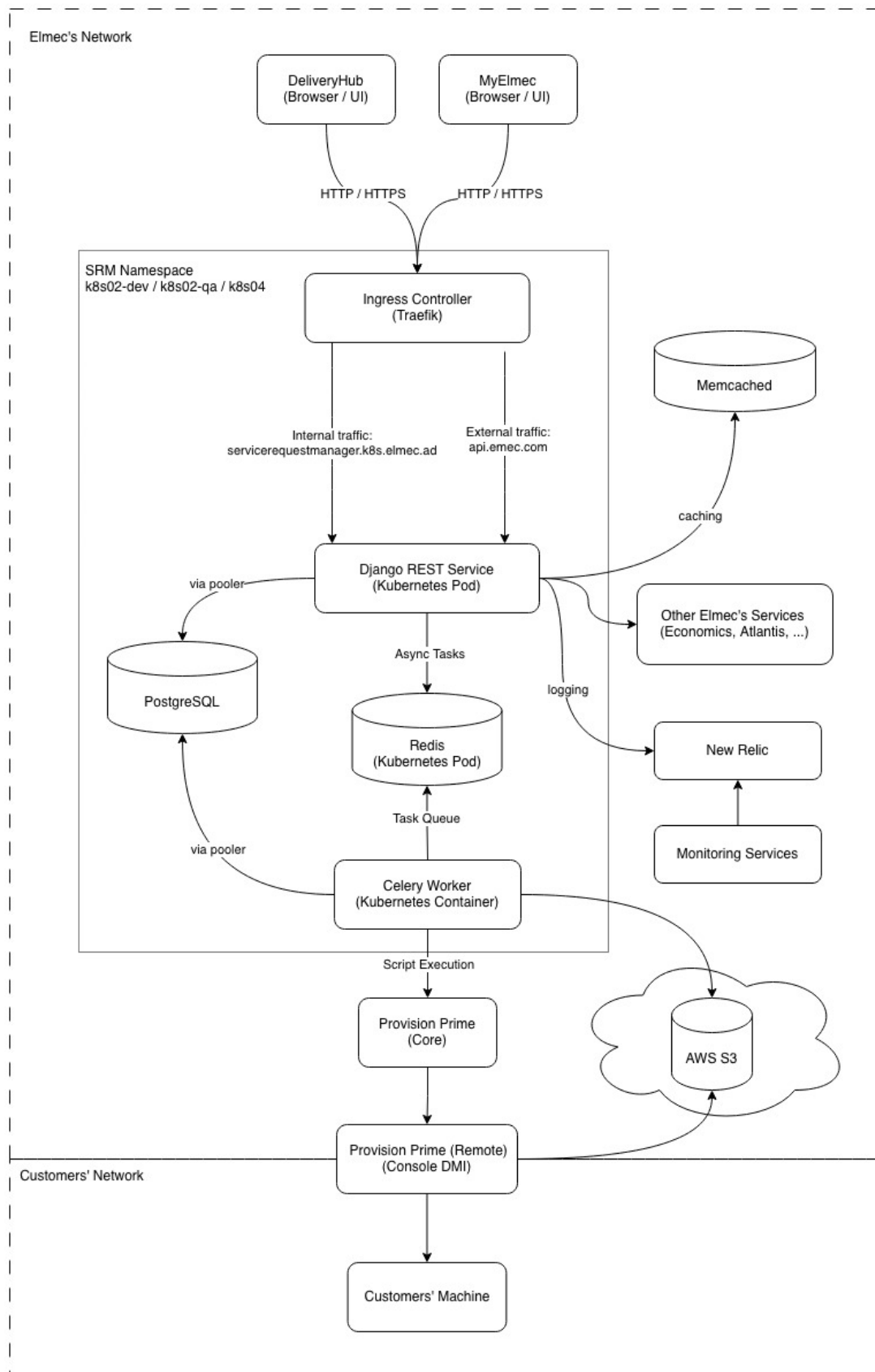


Figure 2.1: SRM System Architecture

REST Services

REST (Representational State Transfer) is an architectural style for designing web services that leverages HTTP protocols for communication between client and server [15]. The basic idea is that a system's resources (users, orders, documents, etc.) are exposed through a uniform interface, typically HTTP, using [15]:

- URLs to identify resources (e.g., `/api/catalog/actions/123/`);
- HTTP verbs to operate on resources (GET, POST, PUT, DELETE, ...);
- Representations of resources in standard formats (JSON, XML, ...).

These services will be responsible for business logic, user request management, and interaction with databases and other system components.

Authentication and Authorization System

Authentication and authorization will be managed through the internally developed library, PyElmecAuth or GoElmecAuth (depending on the implementation language), which will interface with the corporate identity management system to ensure secure and controlled access to the REST services.

It will therefore be necessary to integrate a granular permission system into SRM to ensure that users can only access the resources and functionalities for which they are authorized, based on the roles defined by the corporate Identity Provider.

Caching System

A caching system will be required to improve system performance and reduce the load on the databases.

In particular, responses from calls to external services will be cached, such as third-party APIs needed to validate some fields, or calls to retrieve customer information, to reduce response times and improve the user experience. The time-to-live (TTL) of the caches must be configurable based on the specific needs of each data type.

Message Broker and Task Queue

To manage asynchronous operations and improve system scalability, a Message Broker (e.g., RabbitMQ, Apache Kafka, Redis, ...) will be used in combination with a Task Queue (e.g., Celery).

This is necessary as the SR lifecycle is very complex and involves many operations that can require long execution times [27], such as sending emails,

maintaining HDA³ tickets, integration with external systems, etc. By using a messaging and task queue system, these operations can be executed in the background, without blocking the application's main flow [38].

Provisioning System

It will also be necessary to integrate a provisioning system to execute the scripts related to SRs and those for master data import directly within the clients' networks. This system must be secure, reliable, and scalable, in order to be able to manage a large number of provisioning requests simultaneously.

Elmec already has its own provisioning system called *Provision Prime* [24], which will be integrated with the new SRM system to execute scripts efficiently and securely.

Monitoring and Logging System

To ensure the stability and reliability of the system, a monitoring and logging system will be required. This system will allow tracking system performance, identifying any problems, and analyzing logs to resolve bugs. Tools such as Prometheus, Grafana, New Relic, or other similar tools will be used to implement this system.

Both system metrics (CPU, memory, database usage, API response times, etc.) and application logs (errors, warnings, debug information, etc.) will be tracked, in order to always have a complete view of the system status, and to have the possibility of launching automatic alarms in case of problems (e.g., if the number of responses resulting in 500 (Internal Server Error) in the last hour is $\geq 1\%$).

Containerization and Container Orchestration System

To facilitate system deployment, scalability, and management, a containerization system (Docker) will be used in combination with a container orchestration system (Kubernetes).

This will allow isolating the different system components, simplifying the deployment process, and ensuring greater resilience and scalability of the system, through functionalities such as replicas, cronjobs, automatic scaling of pods, and so on [19].

³HDA (Help Desk Advanced) is a web-based ticketing and service management system developed by Zucchetti. It enables organizations to manage tickets, incidents, and support processes through a centralized platform [17].

2.3 Addressing the Identified Challenges

All the challenges described in section 1.3 have been analyzed, addressed, and resolved through a series of targeted interventions. These interventions concerned various aspects of the system, from restructuring the software architecture and revising data management processes to implementing new technologies and development practices. The solutions adopted for each of the identified challenges are detailed below.

2.3.1 Architectural and Technological Challenges

The adoption of an approach based on containerization (with Docker) and orchestration (with Kubernetes, often abbreviated as K8s) forms the foundation for overcoming the current architectural and technological limitations of the SRP system. This transition not only directly addresses issues of modularity, isolation, and dependency management but also paves the way for a more modern and scalable architecture, such as microservices or micro-frontends. Specifically, it aims to solve the problems of fragmentation, dependency complexity, and maintenance difficulties inherited from the previous monolithic architecture and the agent installed at the customer's site.

Containerization with Docker

Docker is a containerization platform that allows developers to package applications and all their dependencies (libraries, configurations, environments) into a single, standardized unit called a container. As illustrated in figure 2.2, these containers are isolated from each other but share the host operating system's kernel, making them extremely lightweight and portable.

The use of Docker resolves the "works on my machine but not in production" problem (known as "dependency hell") [3] and provides a solid foundation for component decoupling. Each component (frontend, backend, support services) will be enclosed in a *Docker container*, and ensures that the execution environment (including libraries, configurations, and dependencies) is identical across development, testing, and production [3]. This eliminates errors caused by environmental discrepancies.

Once created, the container can be run anywhere Docker is present (in our case, container orchestration is managed by Kubernetes, which will be discussed in the next subsection), simplifying the deployment process. The system thus becomes extremely more portable, moving away from the strict dependence on specific operating systems and hardware configurations like, in this case, Windows Server and Microsoft SQL Server.

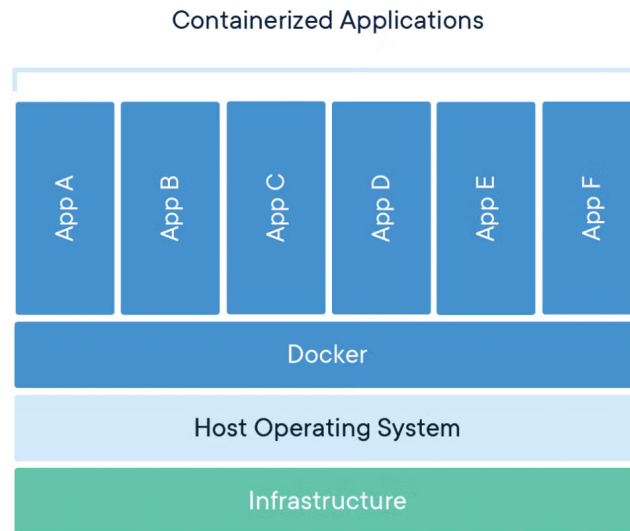


Figure 2.2: Docker Container Architecture (source: [44])

Orchestration with Kubernetes

Kubernetes (K8s) is an open-source system for automated container orchestration, originally developed by Google. Its purpose is to manage entire fleets of Docker containers in a *cluster*, ensuring that applications are always available, scalable, and resilient [19].

If Docker provides the mechanism for creating and distributing containers, Kubernetes is its natural complement, handling their large-scale management (*orchestration*), transforming the infrastructure from a set of single machines into a centrally managed cluster [19].

K8s allows components to scale horizontally based on the workload (via the Horizontal Pod Autoscaler). When demand increases, K8s automatically launches new instances of services (e.g., more pods for the REST backend) and removes them when the load decreases. This is crucial for managing traffic peaks [26].

In parallel, it constantly monitors the status of the containers. If a component fails, K8s automatically restarts it or schedules a new one on a healthy node in the cluster, ensuring high availability and resolving the single-point-of-failure problem introduced by the current dependence on single servers and the agent [26].

Kubernetes also supports advanced deployment strategies (e.g., Rolling Updates), allowing code updates without service interruptions (zero-downtime deployment) [26]. In case of issues, rolling back to the previous version is fast

and automated. This mitigates the risk associated with system changes.

Finally, K8s also offers native mechanisms to manage environment variables and sensitive data (such as database credentials or external service authentication tokens) separately from the container code [26]. This centralizes management and increases security, eliminating the numerous hard-coded configurations present in the old system and making the containers more generic.

The adoption of Docker and Kubernetes therefore creates the ideal environment for the subsequent decomposition of the frontend monolith and the re-engineering of the various backend components, topics that will be covered in detail in the following sections.

Monolithinc Frontend Component

At a technological level, the chosen technology did not change compared to the existing system, which is Vue.js. However, Vue.js version 3 was used, which introduces numerous improvements compared to version 2 used in the existing system, including better performance management, a more flexible composition system, and improved support for TypeScript [43]. In fact, in addition to the updated version of Vue.js, it was decided to adopt TypeScript as the development language for the frontend, in order to improve code maintainability and quality [40].

TypeScript Developed in 2012 by Microsoft as free and open-source software (Apache License 2.0) [40], TypeScript extends JavaScript’s functionalities by introducing optional static typing. The main advantage is that, unlike JavaScript, where type errors are detected only at runtime, TypeScript allows most errors to be caught already at the *build time* stage. This ensures greater code robustness and a significant reduction in production bugs. For extended projects like this one (the Service Requests component is actually only a part of a much larger customer-dedicated platform called *MyElmec*), static typing significantly improves code maintainability and readability, acting as a form of implicit documentation and allowing for safer and faster refactoring.

To solve the maintainability and scalability problems of the existing monolithic frontend, it was decided to move all the configuration and customization logic for field behavior (autofill, dynamic fields via external APIs, validation, dependencies, etc.) to the backend (a topic that will be explored in section 2.3.1). In this way, the frontend was transformed into a simple interface for display and user interaction (as it should be), significantly reducing code complexity and facilitating the implementation of new functionalities. In particular, the frontend is now limited to reading a JSON configuration provided by the backend, which describes the form structure, the fields to be displayed, their properties, and validation rules. This was also made possible thanks to the

potential offered by Vue.js 3, which allows for the creation of highly dynamic and reusable components based on external data [43].

This led to a transition from a total of 86 different "types" of fields in the existing system to only 10 generic field types:

- Text
- Numeric
- File
- Password
- Checkbox
- Select (with already defined options)
- Select (with API-fetched options)
- Date / DateTime
- UPN (managed specifically due to its peculiarities, as it is composed of two sub-fields and has a particular validation logic)
- ADOU Tree

All the various customization logics (especially the customer-specific ones) thus become transparent to the frontend, which merely interprets the configuration provided by the backend. The system transitioned from a component with approximately 11,000 lines of code to a component with only 1,000 lines of code, resulting in improved maintainability and ease of system extension, and with the possibility of implementing other frontends (e.g., a mobile app) with minimal effort without having to replicate all the complex field management logic.

Overly Generic Backend REST Services

To solve the issues related to the frontend, a deep restructuring of the backend logic was necessary, especially concerning fields, moving from JSON responses of this type

```
1 {  
2     "idAction": 455,  
3     "idField": 1649,  
4     "descField": "Manager",  
5     "type": "ADUser",  
6     "idCustomer": 42,  
7     "orderN": 53,
```

```

8     "mandatory": false
9 }

```

to much more detailed and specific responses, which provide the frontend with all the necessary information for the complete rendering of the entire form, such as the following:

```

1 {
2     "field": {
3         "id": 3001,
4         "name": "manager",
5         "description": "Manager",
6         "type": "select",
7         "is_sensitive": false,
8         "option_source_policy": "EXTERNAL_API",
9         "external_api_config": {
10             "name": "AD_USERS",
11             "base_url": "https://q-api.elmec.com/srp/",
12             "endpoint_pattern":
13                 ↪ "api/customer/VFG/adusers?domain={{domain}}",
14             "http_method": "GET",
15             "headers_json": {
16                 "Authorization": "{{TOKEN_TYPE}} {{TOKEN}}"
17             },
18             "response_mapping_json": {
19                 "name": "userDesc",
20                 "value": "user"
21             },
22             "timeout_seconds": 10,
23             "is_searchable": true,
24             "search_param_name": "search",
25             "root_parent_id": null
26         },
27         "options": null,
28         "multiple_choice": false,
29         "file_accept_mimes": null
30     },
31     "mandatory": false,
32     "validation_regex": "^[a-zA-Z0-9._]{5,20}$",
33     "validation_regex_message": "Invalid format for Manager
34     ↪ field.",
35     "autofill_template": null,
36     "depends_on": [
37         "domain",
38         "first_name",
39         "last_name"

```

```
38     ],  
39     "display_order": 26  
40 }
```

In addition to modeling all possible properties of a field in a clear and structured way, the new REST services are also responsible for managing all the customization and configuration logic of the fields, with a three-level structure (which will be detailed in the chapter, dedicated to implementation aspects):

1. Fields
2. Actions
3. Customers

Broadly speaking, the idea is that each field (first level "Fields") has a set of generic properties (type, description, options, etc.) that can be further customized based on the specific action (second level "Actions") to which the field belongs (e.g., making it mandatory or not, adding validation rules, autofill logic, dependencies on other fields, etc.). Finally, each action can be further customized per customer (third level "Customers"), allowing the field's behavior to be adapted to the specific needs of each customer (e.g., modifying the available options in a selection field, changing validation rules, etc.), all without affecting the "original" configuration of the underlying level.

This three-level structure allows for a high degree of reusability and modularity, making it possible to define generic fields that can be easily adapted to different situations without having to duplicate code or logic.

Dependency on the SRP Agent

The dependency on the agent installed at the customer site has been completely eliminated thanks to the integration of Provision Prime, an internally developed provisioning tool written in Go, which manages the execution of Ansible playbooks on remote machines, in this case, the servers within customer networks, transparently to the developer (who will simply configure the playbook and make an API call).

Ansible Ansible is an open-source, agentless IT automation engine used for configuration management, application deployment, intra-service orchestration, and provisioning [4]. It is designed to be simple, powerful, and easy to use, operating over standard SSH for Linux/Unix and WinRM for Windows, thus requiring no special client software (agents) to be installed on the managed nodes [4]. This reliance on existing transport protocols and its human-readable structure, written in YAML, contribute to its minimal learning curve and rapid adoption. The core of Ansible's functionality are

Playbooks. A Playbook is essentially a blueprint of automation tasks, structured as a YAML file, that describes a set of steps to be executed on specified hosts or groups of hosts (defined in an inventory) [4]. They define the desired state of a system in a clear, declarative manner. Each Playbook consists of one or more plays, and each play is an ordered list of tasks. A task is a call to an Ansible module, which is the unit of code that performs a specific action, such as installing a package, copying a folder, or executing scripts. Playbooks ensure that complex, multi-tier application deployments and job executions are repeatable and reusable, transforming manual, error-prone processes into reliable, version-controlled automation.

In this way, not only is the dependency on the agent, and all the associated logic, eliminated, but all the scripts related to the Service Requests, which previously resided physically inside the customers' Domain Controllers and were not tracked in any way, are now versioned.

The cost of these improvements is represented by the additional logic needed to manage communications with Provision Prime, specifically the retrieval of the target machine. To implement these retrieval logics, which are highly custom for each type of target machine (e.g., Domain Controller, Exchange Server, internal workers, etc.), the *Strategy* design pattern [25] was leveraged. This pattern allows each retrieval logic to be encapsulated in a separate class, making the code more modular, extensible, and maintainable (a topic that will be explored in depth in Section ??).

As for the script execution logs, which are needed to monitor the status of Service Requests and detect execution errors, an integration with an AWS S3 bucket, a scalable and durable storage service, will be implemented. This way, every time a script is executed on a remote machine, the logs will be uploaded by the DMI (via the Ansible playbook) to the S3 bucket, ensuring centralized and secure access to execution logs regardless of the machine on which they were generated.

2.3.2 Code and Data Management Issues

The lack of versioning and modern debugging capabilities outlined in the existing system is fundamentally addressed by centralizing all disparate logic (previously scattered across Stored Procedures and SSIS jobs) into a unified, version-controlled software project integrated with a robust CI/CD pipeline. By adopting Git as the standard Version Control System (VCS), all business logic is now treated as application code. This immediately solves the versioning problem: every change is tracked, attributed to an author, and can be reverted, enabling seamless collaboration and a full audit trail. Moreover, moving the core business logic out of the database and into a dedicated backend service (Django REST Framework) written in a modern language (Python) drastically improves debuggability, allowing developers to use powerful, standardized IDE

tools to set breakpoints, inspect variables, and step through code without impacting the production database.

This strategy is completed by integrating several specific CI/CD pipelines (a topic that will be explored in depth in Section 3.1) tailored for each environment branch (**feature**, **devel**, **quality**, **main**). This setup automates the deployment process: code changes are first pushed to a dedicated task branch, reviewed, and then merged into the **devel** environment. Upon passing, the code is merged to the **quality** environment, and finally to **main** for production release. This process ensures that no logic is directly modified in a production environment, eliminating the single-point-of-failure inherent in direct database and server access. Furthermore, replacing the inaccessible, unversioned, and tightly coupled SSIS Jobs with modern, containerized async tasks (managed as code within the Git repository) breaks the strong infrastructure coupling, enhancing accessibility, facilitating collaboration, and making maintenance and debugging a safe, environment-specific activity.

2.3.3 Database Schema and Data Integrity Issues

The issues related to database design and data management were addressed through a series of interventions aimed at improving data integrity, consistency, and maintainability. These interventions include a complete revision of the database schema to eliminate redundancies and inconsistencies, the implementation of stricter integrity constraints, and the adoption of clear and consistent naming conventions for tables and fields.

The system transitioned from using Microsoft SQL Server as the Database Management System (DBMS) to PostgreSQL, an open-source DBMS known for its robustness, scalability, and compliance with SQL standards. PostgreSQL offers advanced features such as support for complex data types, full ACID transactions, and an extensibility system that allows new functionalities to be added via extensions [36]. This choice not only improves the system's performance and reliability but also facilitates data management and the implementation of database design best practices.

Lack of Data Integrity and Consistency

To resolve data integrity and consistency issues, stricter referential integrity constraints were implemented using well-defined primary and foreign keys, also specifying behaviors in case of deletion (**ON DELETE CASCADE** / **SET_NULL**, etc...).

Specifically, for the deletion of "core" entities (such as customers, actions, or fields), the **CASCADE** behavior was chosen for "child" entities, which are consequently deleted.

For secondary entities and/or those that can be null (such as ticket categorization information, action categories and groups, order information associated with tickets, etc...), the `SET_NULL` behavior was chosen to preserve the history of operations and associated data, preventing the loss of more important information.

These constraints ensure that relationships between tables are correctly maintained, preventing the creation of orphaned or inconsistent data.

Data Redundancy

New relationships were created to maximize data reuse and reduce redundancy, while ensuring referential integrity through the use of well-defined primary and foreign keys.

These new entities primarily model concepts that were previously duplicated across multiple tables, such as cost centers (CDC, "*centro di costo*"), service codes, and ticket classification and category. This significantly reduced the amount of redundant data, improving storage efficiency and database maintainability.

Unclear and Inconsistent Naming Conventions

Now, all tables and their attributes follow standardized naming conventions (*snake_case*), improving the readability and maintainability of the database schema.

For "core" relationship names, the plural form was chosen (e.g., `actions`, `service_requests`, `customers`, etc...), while for "child" relationships, the singular name of the "parent" relationship followed by the specific name in the plural was chosen (e.g., `action_fields`, `customer_actions`, etc...).

2.3.4 Maintenance and Operational Challenges

With the new improvements introduced in this chapter concerning architecture, technology, the completely redesigned catalog and configuration management, and the new execution flow, many of the previously encountered maintenance and operational problems have been resolved. The standardization of development and deployment processes, the adoption of DevOps practices, and the implementation of a more modern and scalable infrastructure have led to a significant reduction in downtime, while improving the system's overall reliability. Furthermore, the adoption of advanced monitoring and logging tools allows for rapid identification and resolution of any issues, ensuring smooth and continuous system operation.

These improvements not only facilitate daily operations but also long-term

planning, enabling the system to adapt easily to future needs and challenges.

Chapter 3

Implementation of a modular and scalable solution

While the previous chapter described in detail the requirements and the new architectural design for the new ServiceRequestManager (SRM) system, proposing targeted solutions to overcome the various technological and structural limitations of the legacy SRP system, this chapter marks the transition from the theoretical plan to practical realization, focusing on the concrete implementation of a modular and scalable solution.

The primary focus of this implementation lies in how the critical challenges described in the previous chapters were resolved. Specifically, it will be highlighted how the centralization of field management and configuration logic (and more generally of the catalog) on the backend eliminated the monolithic frontend and enabled a highly reusable three-level configuration system. The integration of the new execution and retrieval flow for master data through a message queue system will also be illustrated, which is essential for eliminating the dependency on the obsolete SRP Agent.

3.1 Automation of the Development Lifecycle through CI/CD Integration

For each feature or significant code modification, a dedicated branch is created named with the ID of the ticket associated with the modification (e.g., IN-7835). These branches are used for isolated development of new features, allowing teams to work in parallel without interference. Once the modification is complete, the branch is submitted to code review and automated testing through the CI/CD pipeline before being integrated into the main branch.

There are 3 distinct live environments: `devel`, `quality`, and `production`. The company Git workflow requires that feature branches be first merged into `devel` for further testing and integration, and subsequently into `quality` for final validation before production release. This approach ensures that each modification is accurately verified and tested in environments progressively closer to production, reducing the risk of introducing bugs or regressions into

the live system.

3.1.1 Feat Branches

For feat branches, the CI/CD pipeline is designed to execute a series of automatic checks aimed at ensuring code quality, security, and compliance before its integration into the main branches, thus not including the build, test, and deploy phases of the application, as shown in figure 3.1.

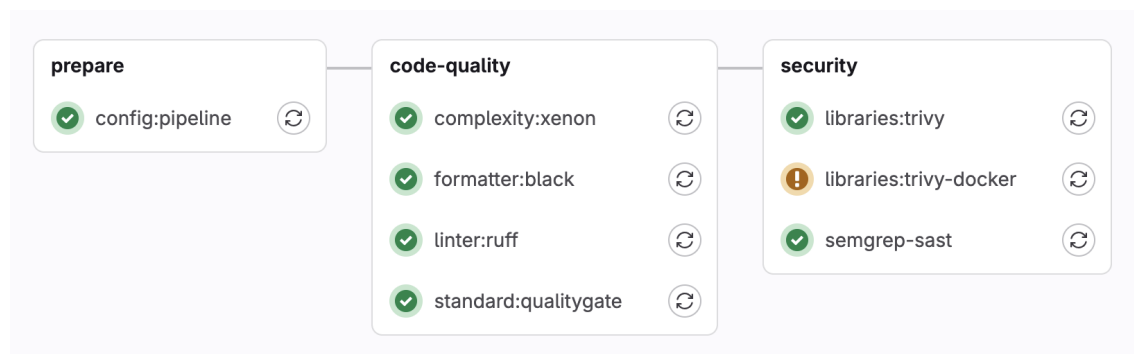


Figure 3.1: Feat Branches pipeline workflow

Prepare

The `config:pipeline` stage is responsible for preparing the execution environment by dynamically retrieving the project configuration from an external service. After installing the necessary tools, it executes an HTTP request to obtain the configuration parameters in JSON format and converts them into standardized environment variables, saving them in a `.env` file. This file is then published as an *artifact*, so that subsequent pipeline steps can use this information to adapt their behavior to the project-specific settings.

Code Quality

Complexity Check The `complexity:xenon` stage is responsible for measuring code complexity through the *Xenon* static analyzer ¹, with the goal of identifying files or functions that might be difficult to maintain. After preparing the Python environment and installing the necessary tool, the job executes the analysis by comparing the results with a predefined quality threshold. The outcome is saved in a report file and sent to the project quality monitoring system. If critical issues that exceed the allowed level are detected, the job signals

¹Radon is a Python static-analysis tool that evaluates code complexity by computing metrics such as cyclomatic complexity, maintainability index, and raw code statistics. It provides a quantitative assessment of the structural quality of Python code. Xenon is a supervisory tool built on top of Radon that enforces predefined complexity thresholds [46].

a failure, thus contributing to keeping software complexity within sustainable limits over time.

Code Formatter The `formatter:black` stage aims to verify that Python code complies with the formatting conventions established by the project (PEP-8) [21], using the *Black* tool. During the job, the shared formatting configuration is downloaded, then a check is executed on the entire repository without applying automatic modifications. The analysis outcome is saved in a report and recorded in the quality monitoring system. If files that do not conform to the required style are found, the job signals an error to ensure that all modifications respect uniform code formatting standards.

Code Linter The `linter:ruff` stage is dedicated to static code checking in order to identify errors, unsafe constructs, or constructs not compliant with internal development guidelines. A Python linter (*Ruff*) is executed that analyzes the entire project and produces a report with any violations or refactoring warnings. The results are collected and sent to the quality monitoring system, contributing to maintaining a uniform style and preventing potential runtime problems. In case of errors considered blocking according to the project-defined rules, the job fails, preventing the integration of non-compliant code.

Quality Gate The `standard:qualitygate` stage statically analyzes the project through a set of rules that verify compliance with company standards. The system inspects Python files, YAML configurations, requirements, and deployment scripts, checking aspects such as proper debug configuration, presence of monitoring systems (*Sensu* / *Prometheus*), appropriate use of standard company libraries, APM (Application Performance Monitoring) configuration, and secure environment variable management. Each rule violation is reported with details about the file and the affected line, blocking the pipeline if critical errors are detected. This approach ensures that only code compliant with security and quality standards can be deployed to production, significantly reducing the risk of operational problems and vulnerabilities.

Security

Libraries The `libraries:trivy` stage is focused on dependency analysis in order to identify known vulnerabilities in packages used by the project. Trivy is used, a tool specialized in scanning libraries and software components against updated CVE² databases [39]. The check result allows for timely identification

²CVE (Common Vulnerabilities and Exposures) are standardized identifiers assigned to publicly known cybersecurity vulnerabilities [10]. Each CVE entry provides a unique ID and

of security risks related to versions affected by known bugs or flaws [39], thus contributing to maintaining an adequate security level and preventing the introduction of vulnerable dependencies in the release process.

The `libraries:trivy-docker` stage, instead, focuses on searching for misconfigurations in deployment files, Dockerfiles, and Kubernetes manifests. Trivy executes a complete scan using a "*comprehensive*" detection priority and generates a detailed report in JSON format [39], which is saved as an artifact to allow subsequent analysis. The scan also verifies dependency licenses and fails if critical unresolved vulnerabilities or significant misconfigurations are detected.

These two stages provide essential feedback on the project's security posture, allowing the team to proactively identify and correct potential security problems before deployment to production.

Static Application Security Testing (SAST) This stage executes static source code analysis to identify security vulnerabilities and potentially dangerous code patterns. *Semgrep* analyzes the code without executing it, detecting issues such as SQL injection, XSS, insecure credential management, and other OWASP vulnerabilities [30]. The stage generates a standardized report in JSON format that classifies vulnerabilities by severity (Critical, Medium, Info) and includes metadata such as commit SHA, branch, and project name. For main branches (quality, devel, master), the results are sent to a centralized project evaluation system for historical tracking. In this way, it is ensured that no code with known security problems can reach production environments.

3.1.2 Live Environment Branches

For live environments, namely those of devel, quality, and production, the CI/CD pipeline includes additional specific phases for building, testing, and deploying the application, as shown in figure 3.2, excluding the phases already covered for feat branches.

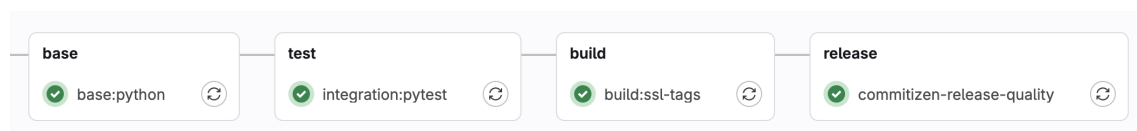


Figure 3.2: Live Environment Branches pipeline workflow (excluding already covered stages)

a brief description of the flaw, enabling consistent referencing across security tools, research, and remediation processes [10].

Base Image Build

The `base:python` stage is responsible for building the project's base Docker image, containing all Python dependencies defined in the `requirements.txt` file.

This stage implements an optimized build strategy through layer separation: the image is rebuilt only when changes are detected in the `requirements.txt` or `Dockerfile_base` files, avoiding unnecessary rebuilds and significantly reducing pipeline execution times. The job is executed exclusively on main branches (devel, quality, master).

This separation into three distinct Dockerfiles (`Dockerfile`, `Dockerfile_base`, `Dockerfile_ssl`) allows for intelligent use of Docker cache: if dependencies don't change, subsequent stages can reuse the existing base image, significantly speeding up the build process and reducing the load on CI/CD runners.

Test

The `test:pytest` stage is responsible for executing the project's automated test suite using PyTest and PostgreSQL as the reference database. The job configures an isolated test environment by starting a dedicated instance of PostgreSQL 16 as a containerized service, with credentials and connection parameters dynamically defined through environment variables. Test execution occurs only if the `requirements_tests.txt` file is present and is adapted based on context: for devel and quality branches, the `devel-latest` and `quality-latest` Docker images are used respectively, while for merge requests the `production-latest` image is employed, thus ensuring that tests are executed in conditions as similar as possible to the target environment.

The stage also implements rigorous control over code coverage: if the coverage percentage is less than 80%, the pipeline automatically fails, ensuring that a minimum level of software quality and reliability is maintained through adequate test coverage. The stage is excluded from tags and the master branch to avoid redundant executions.

This approach ensures that every code modification is validated against a real database before integration, reducing the risk of regressions and compatibility problems.

SSL Image Build

The `build:ssl-tags` stage is dedicated to building the intermediate Docker image containing SSL certificates and security configurations necessary for the project's encrypted communications. This stage is part of the multi-layer build strategy and is executed exclusively on quality and master branches, only if

the `Dockerfile_ssl` file is present. Similarly to the `base:python` stage, it implements a conditional rebuild system that reconstructs the image only when changes are detected in the `requirements.txt` or `Dockerfile_ssl` files, optimizing pipeline execution times through Docker cache reuse. This separation of the SSL layer allows for independent management of security configurations and certificates, facilitating updates and maintenance of encryption-related aspects without having to rebuild the entire application image.

Release

The `commitizen-release` stage manages the automatic versioning process and project release creation using *Commitizen*, a tool that analyzes commit history to automatically determine the version number according to Semantic Versioning³ and Conventional Commit⁴ conventions. The `commitizen-release` stage operates on the master branch and, after configuring Git credentials and synchronizing the local repository with the remote one, executes an automatic version bump based on conventional commit messages (feat, fix, core, refactor, test, ...). The process automatically creates corresponding Git tags and pushes them to the repository with the `ci.skip` option to avoid recursive pipeline activation. The `commitizen-release-quality` stage works similarly but is dedicated to the quality branch and generates pre-release versions with "alpha" suffix, dynamically modifying the Commitizen configuration to adapt it to the test context. Both stages are executed only if the `.cz.yaml` configuration file is present and are excluded from merge requests, ensuring that versioning occurs exclusively on main branches after code integration.

This automated approach eliminates the need to manually manage version numbers and ensures consistency and traceability in the software release process.

3.1.3 Tag Branches

Project Build

The `build:project-tags` stage represents the final phase of the build process, responsible for building the final Docker image of the application ready

³Semantic versioning is a versioning convention that adopts the MAJOR.MINOR.PATCH scheme, where the MAJOR number is incremented in case of changes incompatible with previous versions, the MINOR number following the introduction of compatible features, and the PATCH number for corrective interventions that do not alter existing functionalities.

⁴Conventional Commits is a convention for writing commit messages that uses a standardized structure composed of a modification type (feat, fix, chore, test, refactor, ...), an optional scope, and a description, in order to make change tracking, release note generation, and software versioning clearer and more automatable.

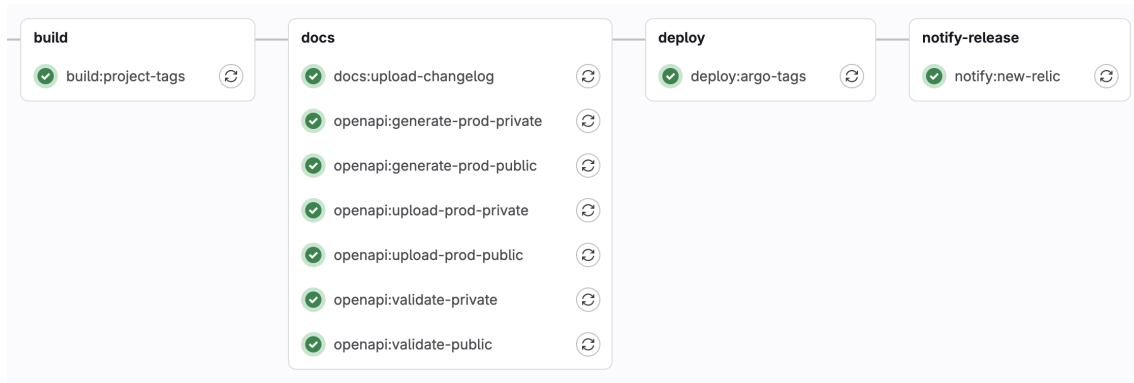


Figure 3.3: Tag Branches pipeline workflow (excluding already covered stages)

for deployment. This stage is activated exclusively when a Git tag respecting the Semantic Versioning format (major.minor.patch) is created, ensuring that only formally released versions are containerized. The job uses the main **Dockerfile**, which assembles all previously built layers (base and SSL), adding the application code and final configurations. The stage implements conditional logic that prevents execution if a separate **Dockerfile_base** exists, delegating in that case the build to more optimized flows that leverage multi-layer cache. The resulting image is tagged with the specific release version and published to the company Docker registry, constituting the definitive artifact ready to be deployed to production environments through subsequent pipeline stages.

Docs

The documentation stages automate the generation and publication of the project's technical documentation. The **docs:upload-changelog** stage is responsible for publishing the changelog on Techpedia, a company collaborative documentation platform, using the GraphQL API to automatically create or update changelog pages when a new release tag is created. The **openapi:generate** stages instead handle the generation of API documentation in OpenAPI/Swagger format, using the Django Spectacular framework to extract the specification from code annotations. These stages produce two versions of the documentation: a public one, containing only externally accessible endpoints, and a private one with complete documentation of all internal APIs. Generation occurs for each environment (production and quality) by connecting to the respective databases to ensure that the documentation accurately reflects the actual data structure. The generated OpenAPI schemas are saved as artifacts and can subsequently be used for automatic validation, SDK client generation, or publication on API documentation portals, ensuring that technical documentation is always synchronized with the code and accessible to developers and API consumers.

Deploy with ArgoCD

The `deploy:argo-tags` and `deploy:argo-tags-quality` stages manage the automatic deployment of the application on Kubernetes environments through ArgoCD⁵. The `deploy:argo-tags` stage is activated exclusively for stable release tags (major.minor.patch format) and performs deployment to the production environment, while `deploy:argo-tags-quality` handles pre-release versions (with alpha, beta, rc suffixes) by deploying them to the quality environment for integration testing and validation. Both stages use a specialized Docker image containing `candyman`, an internally developed wrapper tool that interacts with ArgoCD to automate the declarative deployment process. The tool receives as parameters the project namespace and the version to be deployed, then automatically updates the Git configurations of Kubernetes manifests, triggering the GitOps process managed by ArgoCD that synchronizes the desired state with the cluster.

This approach ensures fully traceable deployments, simplified rollbacks, and full adherence to Infrastructure as Code principles, eliminating manual interventions and reducing the risk of errors during release.

Notify Release

The `notify:new-relic` stage is responsible for automatically registering new application releases on the *New Relic* Application Performance Monitoring platform, creating a deployment marker that facilitates correlation between software versions and performance metrics. The job is activated for any tag that respects the Semantic Versioning format and uses the New Relic GraphQL API to identify the correct application through its GUID, automatically distinguishing between the production and quality instance based on the presence or absence of pre-release suffixes in the tag. Once the monitored entity is identified, the stage creates a change tracking event that records the deployed version, allowing operators and development teams to visualize in New Relic dashboards when deployments occurred and to correlate any performance anomalies or errors with specific software versions. This automatic notification mechanism eliminates the need for manual deployment registration and provides complete visibility on the application lifecycle, supporting troubleshooting activities, regression analysis, and evaluation of release impact on system stability.

⁵ArgoCD is a GitOps tool for Kubernetes that automatically keeps the state of applications in the cluster synchronized with that declared on Git [5].

3.2 Implementation of the Deployment Architecture on Kubernetes

The SRM deployment architecture on Kubernetes is designed to ensure scalability, resilience, and separation of responsibilities through an organization based on distinct deployments, each optimized for specific system functions.

As mentioned earlier, there are three separate live environments: `devel`, `quality`, and `production`. For each environment, a corresponding deploy file is defined. For simplicity, the production one (`production.yaml`) will be analyzed, knowing that the other files share exactly the same structure, except for the environment variables (which reflect their respective environments) and the resources, which for `devel` and `quality` are lower than their production counterparts, since those environments are used only internally and therefore need to handle exponentially lower workloads compared to the production environment.

The high-level structure of deployment files is as follows:

```
1 namespace: servicerequestmanager
2 serviceAccount:
3   create: TRUE
4 deployments:
5   - name: static
6     ...
7   - name: backend
8     ...
9 jobs:
10  - name: sync-srp-catalog
11    ...
12  - name: sync-srp-sr-history
13    ...
14  - name: import-directory-data
15    ...
```

The system is isolated within the dedicated `servicerequestmanager` namespace, ensuring logical separation of resources and facilitating permission and network policy management. The configuration provides for the creation of a dedicated service account, which allows pods to interact with Kubernetes APIs according to the *least privilege* principle, limiting permissions only to those strictly necessary for system operation.

3.2.1 Static Content Deployment

The `static` deployment manages the distribution of static content through a containerized Nginx web server, configured to serve frontend assets such

as JavaScript bundles, CSS, images, and other static files. This deployment implements an optimized caching strategy that reduces the load on the backend and significantly improves response times for end users.

```
1   - name: static
2     services:
3       - name: static
4         port: 80
5     ingresses:
6       - name: static
7         public: False
8         type: http
9         host: servicerequestmanager.k8s.elmec.ad
10        path: /static
11        port: 80
12    containers:
13      - name: nginx
14        image: docker.elmec.com/.../ssl:production-latest
15        port: 80
16      - name: nginx-mtr-exp
17        image:
18          ↪ docker.elmec.com/.../nginx-prometheus-exporter:0.8.0
19        port: 9113
20        args:
21          - "-nginx.scrape-uri"
22          - "http://127.0.0.1:8080/nginx_status"
23    replicaCount: 1
24    monitoring:
25      - name: nginx-metrics
26        port: 9113
27    imagePullSecrets:
28      - name: docker-registry-secret
```

Container Configuration

The deployment consists of two containers running on the same pod.

The main `nginx` container uses the `ssl:production-latest` image, which includes SSL/TLS configurations and certificates necessary for secure communications. The server is configured to listen on port 80 within the cluster, while external access occurs through the ingress system that manages SSL termination. In parallel, the `nginx-mtr-exp` container runs the *Nginx Prometheus Exporter*, a specialized tool that exposes web server operational metrics in Prometheus-compatible format. This sidecar container connects to the `nginx_status` endpoint exposed by Nginx on port 8080 and translates native statistics into Prometheus metrics accessible on port 9113, allowing

the monitoring system to collect data on throughput, latency, errors, and connection usage.

Service and Ingress Configuration

The `static` Kubernetes service exposes the deployment on port 80, providing a stable access point independent of the lifecycle of underlying pods. The associated ingress is configured as internal (not public) and routes requests coming from the host `servicerequestmanager.k8s.elmec.ad` with path `/static` to the service, implementing virtual hosting that allows segregating static traffic from API traffic.

Monitoring Integration

The deployment natively integrates monitoring through the definition of a *ServiceMonitor* that instructs Prometheus to scrape metrics exposed on port 9113. This configuration allows automatic collection of operational telemetry without the need for additional manual configurations, ensuring continuous visibility on the state of the static content service.

3.2.2 Backend Application Deployment

The `backend` deployment constitutes the core of the architecture and manages the entire application logic through a multi-container organization that separates responsibilities between API server and asynchronous workers.

```
1 - name: backend
2   vault:
3     name: 'secrets'
4   services:
5     - name: api
6       port: 5000
7       mirror: TRUE
8     - name: api-ext
9       port: 5000
10      mirror: True
11   ingresses:
12     - name: api
13       public: False
14       type: http
15       host: servicerequestmanager.k8s.elmec.ad
16       path: /
17       port: 5000
18       middlewares:
19         - name: cors
20           spec:
```

```

21         headers:
22             accessControlAllowMethods:
23                 - "GET"
24                 - "OPTIONS"
25                 - "POST"
26                 - "PUT"
27                 - "DELETE"
28                 - "PATCH"
29             accessControlAllowOriginList:
30                 - "*"
31             accessControlMaxAge: 100
32             accessControlAllowHeaders: ["*"]
33 - name: api-ext
34   public: True
35   type: http
36   host: api.elmec.com
37   path: /servicerequestmanager/
38   port: 5000
39   middlewares:
40     - name: strip
41       spec:
42         replacePathRegex:
43           regex: ^/servicerequestmanager/(.*)
44           replacement: /$1
45     - name: cors
46       spec:
47         headers:
48             accessControlAllowMethods:
49                 - "GET"
50                 - "OPTIONS"
51                 - "POST"
52                 - "PUT"
53                 - "DELETE"
54                 - "PATCH"
55             accessControlAllowOriginList:
56                 - "*"
57             accessControlMaxAge: 100
58             accessControlAllowHeaders: [ "*" ]
59 containers:
60   - name: api
61     ...
62   - name: celery
63     ...
64 replicaCount: 2
65 monitoring:
66   labels:

```

```

67     port: "5000"
68     host: "localhost"
69     subscriptions:
70     - django
71     type: PodAndService
72     podmonitors:
73     - name: django-hc
74       port: api
75       path: /ht/
76       healthCheck: True
77     imagePullSecrets:
78     - name: docker-registry-secret

```

Vault Integration for Secrets Management

The deployment is configured to integrate *HashiCorp Vault*⁶ through automatic injection of secrets into the pod filesystem. The *Vault Agent injector* dynamically mounts secrets (database credentials, API keys, passwords) in the path `/vault/secrets/secrets`, making sensitive information available to containers without exposing them in environment variables or ConfigMaps. This approach implements the secrets management pattern according to Kubernetes security best practices, ensuring automatic credential rotation and complete audit trail of accesses.

API Container Configuration

The `api` container runs the Django application through the Gunicorn server, configured to handle HTTP requests on port 5000. The resource configuration implements a *Burstable Quality of Service*⁷ model, with requests of 400m CPU and 1000Mi memory and limits of 800m CPU and 1200Mi memory respectively. This configuration allows the container to temporarily scale beyond requested resources during load peaks, while ensuring a minimum baseline guaranteed by the Kubernetes scheduler.

⁶HashiCorp Vault is a system for securely managing secrets such as passwords, tokens, certificates, and API keys. It's used to centralize, encrypt, and dynamically deliver these secrets to pods, avoiding storing them in plaintext in YAML manifests or Kubernetes Secrets. It also enables secret rotation, temporary access with fine-grained control, and automatic secret injection into containers via sidecar or the CSI driver.[42]

⁷Burstable Quality of Service (QoS) is a resource classification applied to pods whose requested CPU/memory is lower than their limits. This means the pod is guaranteed at least its requested resources, but it's also allowed to "burst" up to its limits when the node has spare capacity. Burstable sits between Guaranteed (highest priority) and BestEffort (lowest priority) in terms of scheduling and eviction priority [23].

```

1 containers:
2   - name: api
3     image: "docker.elmec.com/innovation/servicerequestmanager:PR
      ↪ PROJECT_VERSION"
4     port: 5000
5     cpuRequest: 400m
6     cpuLimit: 800m
7     memoryRequest: 1000Mi
8     memoryLimit: 1200Mi
9     envs:
10      - name: ENVIRONMENT
11        value: "PRODUCTION"
12      - name: DJANGO_DB_HOST
13        value: 'servicerequestmanager-db-production-pooler-rw.
      ↪ autom-servicerequestmanager-production.svc.cluster
      ↪ .local'
14      - name: DJANGO_DB_PORT
15        value: '5432'
16      ...

```

Environment variables define the application's operational configuration, including PostgreSQL database connection parameters through the PgBouncer pooler, distributed caching configuration via Memcached, integration with Redis for Celery queue management, and external service endpoints for OIDC authentication and integrations with other company systems.

Celery Worker Container

```

1 - name: celery
2   image: "docker.elmec.com/innovation/servicerequestmanager:PR
      ↪ OJECT_VERSION"
3   memoryLimit: 1200Mi
4   memoryRequest: 1000Mi
5   cpuRequest: 400m
6   cpuLimit: 800m
7   command: "/bin/sh"
8   args:
9     - -C
10    - . /vault/secrets/secrets;cd servicerequestmanager;
      ↪ celery -A servicerequestmanager worker --loglevel=info
      ↪ --pool threads
11   envs:
12     - name: CELERY_QUEUE
13       value: "servicerequestmanager_production"
14     - name: REDIS_HOST

```



```

15     value: "autom-servicerequestmanager-redis.autom-servicer_
      ↪     equestmanager-production.svc.cluster.local"
16     ...

```

The `celery` container runs Celery workers dedicated to asynchronous processing of background tasks related to the service request execution flow, using a thread pool to handle concurrent operations without blocking. The custom startup command initializes the environment by loading secrets from Vault before starting the worker with logging configured at INFO level. Workers are configured to consume from the dedicated `servicerequestmanager_production` queue on Redis, ensuring isolation from other systems sharing the same messaging infrastructure. Resource allocation is identical to that of the API container, reflecting similar computational requirements.

Service Exposure and Load Balancing

The deployment exposes two distinct services to handle internal and external traffic.

The `api` service operates internally to the cluster on port 5000, configured with active mirror mode to allow traffic mirroring for testing or debugging purposes. The associated ingress is configured as internal and routes requests coming from the host `servicerequestmanager.k8s.elmec.ad` to the backend, implementing CORS middleware that permits cross-origin requests with complete HTTP methods (GET, POST, PUT, DELETE, PATCH) and arbitrary headers.

The `api-ext` service instead provides public access through the host `api.elmec.com` with path prefix `/servicerequestmanager/`. The ingress configuration includes a path rewriting middleware that removes the prefix before forwarding requests to the backend, allowing the application to operate with simplified routing paths. This ingress also implements permissive CORS policies to support integrations from third-party web applications.

Availability and Scaling

The deployment is configured with `replicaCount: 2`, ensuring redundancy and load distribution across at least two pods distributed on different cluster nodes. This configuration implements basic high availability, allowing rolling updates without downtime and tolerance to single pod or node failures. The Kubernetes scheduler automatically distributes replicas across different nodes when possible, maximizing resilience.

Health Checking and Monitoring

The monitoring system is configured through PodMonitor that combines application metrics and health checks. The `django-hc` PodMonitor executes

periodic checks on the `/ht/` endpoint of the API service, verifying that the application is responsive and correctly configured.

The deployment is also tagged with `django` subscription that activates pre-defined dashboards and alerts for Django applications, providing immediate visibility on metrics such as request rate, latency distribution, error rate, and database connection pool utilization.

3.2.3 Scheduled Jobs

The architecture includes Kubernetes CronJobs for triggering Master Data export from clients' Domain Controllers and Exchange Servers, and for keeping data aligned between ServiceRequestManager and the legacy SRP system, so that the two systems can, at least initially, coexist and remain aligned during this period. The CronJobs share the same environment variables configuration and secrets injection as the backend deployment, ensuring consistency in access to databases and external services.

Here is an example of the CronJob definition configuration:

```
1 jobs:
2   - name: sync-srp-catalog
3     vault:
4       name: 'secrets'
5     schedule: "0 1 * * *"
6     image: docker.elmec.com/innovation/servicerequestmanager:P_
7       ↪ ROJECT_VERSION
8     memoryLimit: 700Mi
9     memoryRequest: 400Mi
10    cpuRequest: 400m
11    cpuLimit: 800m
12    command: "manage.sh"
13    args:
14      - "sync_srp_catalog"
15    envs:
16      ...
17    imagePullSecrets:
18      - name: docker-registry-secret
```

SRP Catalog Synchronization

The `sync-srp-catalog` job is scheduled for daily execution at 1:00 AM through the cron expression `0 1 * * *`⁸. This job executes the Django management command `sync_srp_catalog` which queries the SRP database to import catalog configurations. Execution is configured with more contained resources compared to permanent deployments (400Mi-700Mi memory, 400m-800m CPU), reflecting the batch nature of the workload that can tolerate higher latencies.

Service Request History Synchronization

The `sync-srp-sr-history` job is scheduled for execution at 2:00 AM, one hour after catalog synchronization to ensure that updated configurations are already available. This job executes the `sync_srp_service_requests` command which imports the history of service requests created in the legacy system starting from the date defined in `SRP_SR_SYNC_START_DATE`, keeping the unified view of requests aligned across the two systems during the transition period.

Master Data Import

The `export-master-data` job is responsible for periodically triggering the export of customers' master data from their Domain Controllers and Exchange Servers via an Ansible Playbook. Scheduled to run every day at 02:00 AM, this job executes the `import_master_data` management command.

This execution model based on Kubernetes Jobs ensures automatic retries in case of failure and retention of execution history for debugging and auditing.

3.3 Backend Service Implementation

The backend service is the most critical component of the SRM architecture, responsible for managing application logic, processing service requests, and integrating with external systems. Its implementation was designed to ensure

⁸The *cron* syntax is a compact format used to schedule periodic execution of processes or jobs in Unix-like systems [6]. It consists of five numeric fields separated by spaces (minute, hour, day of month, month, day of week) that indicate when a command should be executed. Each field can contain single values, ranges (e.g., 1-5), lists (e.g., 1,3,7), or special characters such as *, which means "any value", and /, which indicates a frequency (for example */5 for every five time units) [6]. A string like `0 2 * * *` therefore indicates the execution of a job every day at 02:00. This syntax, while minimal, allows for precise and flexible description of virtually any recurring scheduling need.

scalability, resilience, ease of maintenance, and a high degree of customization to accommodate the specific needs of clients.

3.3.1 Technological Stack

The backend leverages several key technologies that form the foundation of the system architecture:

Core Framework

The backend is built on *Django* 5.2 with *Django REST Framework* (DRF) 3.16, implementing a Model-View-Controller pattern where Django models handle data persistence, DRF viewsets manage business logic and request processing, and serializers handle data validation and transformation [12][13]. This architecture was selected over alternatives like Go with GORM primarily due to development velocity considerations: Django’s comprehensive ecosystem of built-in components (ORM, authentication, admin interface) significantly reduces time-to-market compared to implementing equivalent functionality in Go. While Go offers superior raw performance, the anticipated load characteristics of SRM (primarily CRUD operations and orchestration) do not justify trading ecosystem maturity for performance gains. Django’s sophisticated ORM with features like `select_related` and `prefetch_related` for query optimization, comprehensive migration framework, and automatically generated admin interface provided immediate value during development and deployment phases.

Asynchronous Processing

Celery with *Redis* as the message broker implements the asynchronous task queue for service request execution and scheduled operations [8]. This producer-consumer architecture enables Django to publish tasks to Redis while dedicated worker processes consume and execute them independently, preventing long-running operations from blocking API responses. Celery was configured with thread-based workers optimized for I/O-bound operations (API calls, database queries, remote script execution coordination) rather than CPU-intensive processing. The system leverages Celery’s built-in retry mechanisms with exponential backoff and comprehensive task lifecycle monitoring to ensure reliable execution even during transient failures.

Data Layer

PostgreSQL serves as the primary data store, selected for its robust ACID transaction guarantees, sophisticated query planning, and support for partial indexes and Common Table Expressions (CTEs) essential for hierarchical catalog queries. The application connects through *PgBouncer*, a connection pooler

that maintains persistent database connections and enables horizontal scaling of application pods without exhausting database connection limits.

Memcached implements distributed in-memory caching specifically for responses from external financial and administrative systems (contracts, cost centers, project codes). These external services exhibit high latency but return data with temporal stability, making them ideal candidates for caching with configurable TTL values. The cache-aside pattern transparently reduces redundant external calls without requiring modifications to external systems.

Remote Execution

Ansible orchestrates service request execution on customer infrastructure through agentless SSH/WinRM connections, eliminating the dependency on the legacy SRP Agent. Custom playbooks encapsulate provisioning workflows that currently wrap existing PowerShell scripts (providing immediate SRP Agent elimination while preserving proven automation logic) with a migration path toward native Ansible modules. Playbooks are templated with Jinja2, receiving customer-specific parameters from Django and executing remotely through company-internal orchestration services.

Object Storage

S3-compatible object storage (*Ceph RGW* via MinIO API) manages master data exports and execution artifacts. This decouples export operations from database availability, provides natural archival mechanisms with lifecycle policies, and enables atomic updates of complete datasets. Master data is exported to JSON in S3, then materialized in PostgreSQL for efficient querying, achieving sub-second response times compared to multi-second LDAP query latencies while maintaining data freshness through nightly scheduled exports.

3.3.2 Application Architecture

The Django project is organized into six specialized applications following separation of concerns principles:

- **catalog/** - Manages the three-level configuration hierarchy (Actions, Fields, customer/action-specific overrides), validation rules, and API configurations
- **integration/** - Handles external system integrations (HDA ticketing, financial systems, CMDB)
- **execution/** - Implements service request lifecycle management, field validation, and Ansible execution orchestration

- **monitoring/** - Provides health checks for configuration consistency and system integrity
- **directory_data/** - Manages master data import pipeline from S3 to relational database
- **common/** - Contains shared utilities and reusable components

3.3.3 Catalog Management System

The catalog implements a three-level configuration hierarchy that eliminates the monolithic field duplication of the legacy system while providing granular customization capabilities.

Configuration Hierarchy

As shown in Figure 3.4, the architecture consists of:

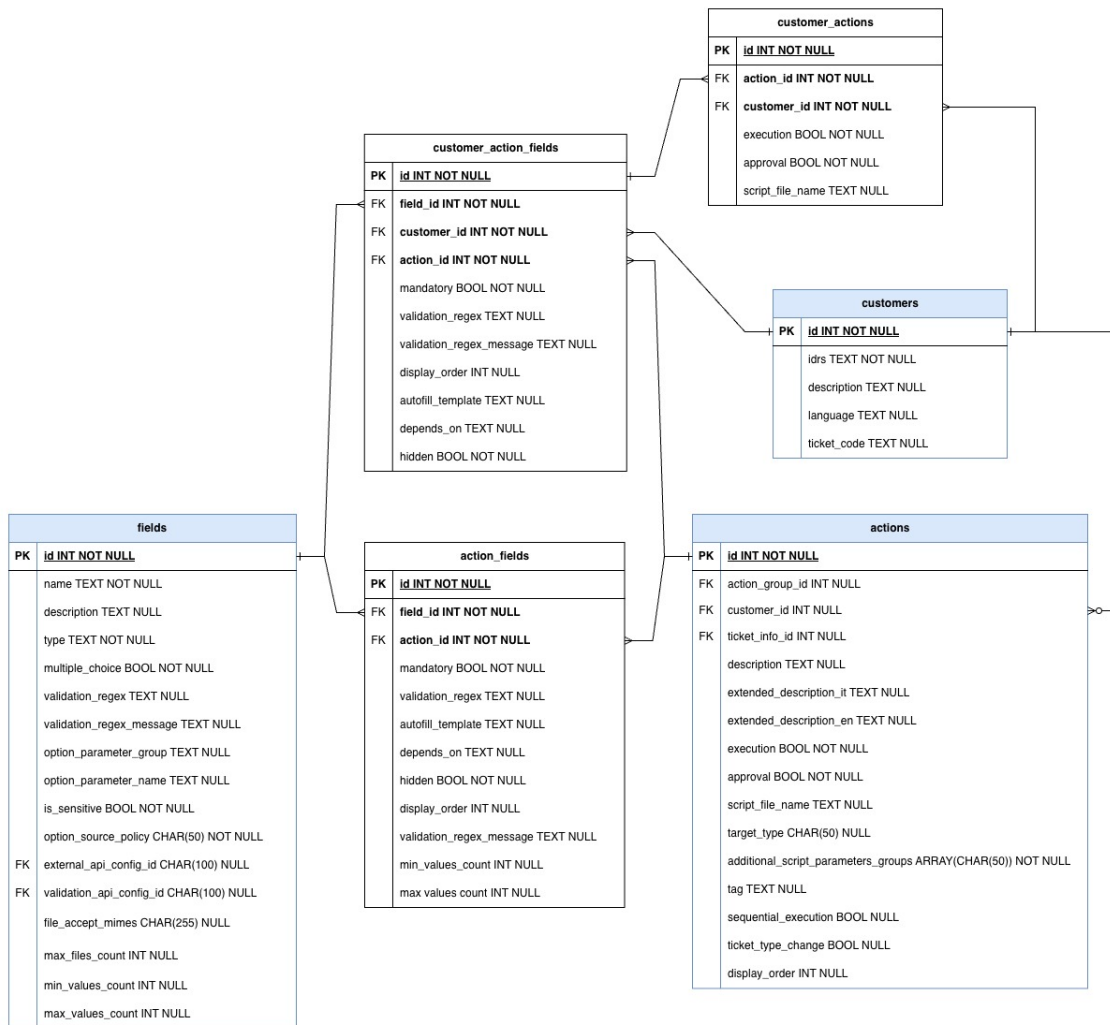


Figure 3.4: ER Diagram of the Three-Level Catalog Configuration Hierarchy

Level 1 - Global Definitions: Base Action and Field models define default

configurations applicable across all customers, including execution scripts, approval requirements, target infrastructure types, validation regex patterns, and data types. The `ActionField` association model links fields to actions while allowing action-specific property overrides.

Level 2 - Customer Action Customization: `CustomerAction` enables customer-specific modifications to action properties (execution scripts, approval workflows, target types) without duplicating the entire action definition, maintaining clear separation between shared logic and customer variations.

Level 3 - Field-Level Customization: `CustomerActionField` provides granular control over individual field behavior for specific customer-action combinations, supporting modification of existing field properties, addition of fields not in the default definition, removal of fields via `hidden=True`, and overrides of action-level customizations.

Configuration resolution follows strict precedence: `CustomerActionField` overrides `ActionField`, which overrides `Field`, while `CustomerAction` overrides `Action`. This is transparently applied by backend retrieval methods, ensuring frontends receive effective configurations without hierarchy awareness.

Key Architectural Patterns

Service Block-Based Action Association: Customer access to actions is determined through service blocks (derived from contract information) rather than granular one-to-one associations. Actions associated with service blocks become available to any customer possessing those blocks, significantly reducing administrative overhead while providing flexible commercial packaging. The system maintains support for direct customer-action associations for edge cases and legacy granular associations during SRP transition.

Field Type System: The catalog implements 14 distinct field types including simple types (text, textarea, password, email, numeric, date/datetime), complex types (text lists with cardinality constraints, single/multi-select drop-downs, UPN composite inputs, AD OU trees, file uploads), and specialized validation for each type. Select fields support three option source policies: customer-specific options (domains, users, databases), globally scoped options (countries, languages), and dynamic options from external APIs configured through `ExternalAPIConfiguration` objects.

Templating System: Jinja2 templating enables dynamic behavior throughout the configuration layer. Field autofill templates define patterns for pre-populating values based on other fields (e.g., `{{ first_name }}.{{ last_name }}`), while external API endpoint patterns incorporate templating for context-specific URL construction (e.g., `/api/customers/{{ idrs`

}}/adusers?domain={{ domain }}). This capability is utilized by both validation logic and frontend presentation, ensuring consistent behavior.

Target Machine Strategy Pattern: Service request execution location is abstracted through the Strategy design pattern. Each action specifies a `target_type` enumeration value mapping to a concrete strategy class implementing `BaseActionTargetType`. Strategies encapsulate logic for identifying target determination fields, validating presence in action field sets, extracting values from service requests, and resolving final target hostnames. This enables addition of new target types without modifying core execution logic.

3.3.4 Core Execution Workflows

Field Validation

The validation system, centralized in `ServiceRequestValidationMixin`, implements comprehensive pre-submission validation that ensures data integrity and provides immediate user feedback. The architecture separates global validators (operating on complete field sets to verify duplicate prevention and mandatory field completeness) from field validators (examining individual field-value pairs for syntactic, semantic, and configuration compliance).

Validation progresses through multiple layers: syntactic validation via regex patterns and type-specific checks, configuration-based validation ensuring submitted fields match the effective catalog configuration, option validation for select/UPN fields against rendered or fetched option sets, and specialized validation for AD OU fields and external validation endpoints. The framework implements error aggregation, continuing validation after detecting failures to accumulate all errors before returning consolidated responses, significantly improving user experience over fail-fast approaches.

Asynchronous Execution Flow

The execution workflow represents a fundamental departure from SRP's polling-based SSIS job mechanism, implementing event-driven task orchestration. Service request creation directly invokes SRM for validation and creation, SRM creates the HDA ticket, and Celery tasks are immediately triggered without polling delays.

The flow implements a state machine (NEW, WAIT4APPROVAL, APPROVED, PENDING, EXECUTED, SUSPENDED, ERROR, REJECTED) coordinated through six specialized Celery tasks that handle new request routing based on approval requirements, approved request execution initiation, rejection and suspension workflows, execution completion processing via webhooks from the orchestration platform, monitoring logic for stalled executions, and asynchronous file attachment handling.

Script execution occurs through Ansible playbooks that implement a five-phase pattern: preparation (create directories on target), file transfer (copy scripts using `win_copy`), execution (invoke PowerShell via `win_shell`), logging (retrieve structured output), and cleanup (remove temporary artifacts). This architecture maintains complete transparency for script developers while providing centralized execution logging and state management.

Master Data Import Pipeline

The master data import subsystem eliminates real-time LDAP query latency through a scheduled export-materialize-query pattern. Nightly CronJobs trigger Ansible playbooks that execute PowerShell queries on customer Domain Controllers to extract users, groups, OUs, password policies, and UPN suffixes in JSON format, upload results to S3-compatible object storage with hierarchical organization, and signal completion for import processing.

Import tasks retrieve JSON files from S3, parse content into typed data structures, and implement sophisticated differential synchronization using Django bulk operations. The system constructs sets of entity identifiers from both imported data and existing database records, performs set operations to determine records to create/update/delete, and executes batched operations (`bulk_create`, `bulk_update`, `filtered delete`) within atomic transactions.

The materialized data is exposed through Django model managers providing filtering and search capabilities for validation logic and frontend autocomplete, achieving sub-second query response times while maintaining freshness through nightly updates.

3.3.5 Observability and Monitoring

The system implements comprehensive observability through multiple layers:

Application Performance Monitoring: New Relic APM provides continuous monitoring of transaction traces, error rates, and resource utilization, instrumented through the `newrelic-admin` wrapper capturing web transaction performance, database query profiling, external service call durations, and Celery task execution metrics.

Structured Logging: An internal library wrapping Loguru provides structured logging with automatic New Relic Logs integration and contextual enrichment (request IDs, user identifiers, customer IDRS, service request IDs). The `@monitor_job` decorator wraps scheduled tasks and management commands, registering them as background transactions, capturing execution metadata, handling exceptions with full tracebacks sent as custom events, and

generating completion events with duration metrics.

Health Checks: The `/ht/` endpoint implements Django Health Check framework with standard infrastructure checks (database connectivity, cache availability, storage accessibility, migration status) and custom application-specific checks. `PendingServiceRequestsCheck` validates that stalled executions have correct ticket assignments, while `UniqueActionFieldsCheck` verifies catalog configuration integrity by detecting duplicate field names within action scopes.

Metrics Exposure: Kubernetes PodMonitors enable Prometheus scraping of application metrics and health check execution results, with predefined Grafana dashboards and alert rules activated through the `django` subscription label.

3.4 Frontend Application Implementation

The SRM frontend comprises two distinct Vue.js 3 applications: an end-user service request creation interface integrated into MyElmec (customer portal), and an administrative configuration platform (currently in UX design phase) for catalog management to be integrated into DeliveryHub (internal operations portal).

3.4.1 Technology Stack

Both applications leverage Vue 3 Composition API with TypeScript for improved code organization and type safety [43]. State management is implemented through Pinia, providing centralized management of authentication state, customer context, form data, and cached external options [22]. The component architecture utilizes a custom design system library (ElmecUI) ensuring consistent styling and behavior across all company applications with accessibility features and responsive design patterns.

3.4.2 Service Request Creation Interface

The end-user interface implements a dynamic form builder that renders creation forms based on effective catalog configuration retrieved from backend APIs. When users navigate to create a service request, a single GET request to `/api/catalog/customers/{IDRS}/actions/{actionId}/` returns complete action definitions including resolved field configurations, validation rules, dependencies, and external API configurations, eliminating frontend configuration resolution logic.

The component architecture separates concerns through a parent `NewServiceRequest.vue` managing overall form state and submission, and

specialized `ServiceRequestField.vue` components rendering individual fields based on their types. The system implements 14 distinct field type renderings including text/textarea with optional autofill template support using Nunjucks for Jinja2 template evaluation, text lists with dynamic array inputs respecting cardinality constraints, select fields with three data source patterns (pre-rendered, external API with asynchronous loading, searchable with remote filtering), UPN composite inputs combining text prefix and domain suffix selection, password fields with confirmation validation, file uploads with MIME type filtering and Base64 encoding, and AD OU tree pickers for organizational unit selection.

Field dependency resolution is implemented through reactive watchers monitoring `formData` changes and evaluating dependency conditions. Fields specify `depends_on` arrays defining prerequisite fields that must be populated before dependent fields become enabled. When dependency values change, the system resets dependent field values, clears cached external options, and triggers re-fetching for newly enabled fields.

Client-side validation leverages Element Plus form validation rules dynamically generated from action field configurations, including required field checks, regex validation with configured patterns, and custom validators for password confirmation. Upon submission, the system executes comprehensive validation, collecting all errors before proceeding, then constructs request payloads containing action and field identifiers with values, transmitting via POST to the backend API.

External API option caching minimizes redundant backend calls through a `calledUrls` Set tracking previously fetched URLs and an `externalOptions` object storing results keyed by API configuration name. Searchable fields implement debounced remote filtering, triggering API calls only after typing pauses.

3.4.3 Administrative Configuration Platform

The administrative interface (in design phase) will provide CRUD operations for managing catalog hierarchy components (Actions, Fields, ActionField associations, CustomerActions, CustomerActionFields). The platform will implement form-based editors for field properties including validation patterns, autofill templates, external API configurations, and dependency definitions, with visual representation of override relationships and configuration conflict detection. Preview functionality will enable administrators to render test forms showing end-user experiences before production deployment, with integration to backend validation endpoints providing real-time configuration correctness feedback.

The frontend architecture achieves the objective of providing dynamic,

configuration-driven interfaces that adapt to catalog changes without code modifications or redeployment, enabling rapid iteration on service request workflows and supporting business agility for managed service operations.

Chapter 4

Coexistence and Gradual Migration Strategy

The transition from the legacy SRP system to the new ServiceRequestManager architecture represents a substantial technical and operational challenge that requires careful planning to minimize business disruption while ensuring data integrity and system reliability. Rather than implementing a disruptive "big bang" migration that would introduce significant risk and potential service interruptions, the project adopts a phased coexistence strategy that enables gradual migration of customers and service request types while maintaining full operational continuity.

4.1 Migration Strategy Overview

The migration architecture implements a dual-flow execution model where both SRP and SRM systems operate simultaneously, with individual service request actions configured to route through either the legacy or modern execution pipeline based on the `target_type` attribute of the Action model. This attribute determines the execution mechanism for each action, with values including Domain Controller, Exchange Server, Internal Worker, API Call, and SRP Agent.

Actions configured with `target_type = SRP` continue routing through the legacy SRP Agent execution mechanism, preserving the existing workflow without modification. Actions configured with any other target type utilize the new Ansible-based execution pipeline through Provision Prime, benefiting from improved logging, error handling, and state management capabilities introduced in SRM.

This per-action migration control provides several strategic advantages: it enables pilot deployments on specific customers or action types, allowing validation of the new architecture in production with controlled scope; it maintains business continuity by ensuring that any discovered issues affect only the subset of actions migrated to the new flow; it supports progressive learning, enabling the operations team to develop expertise with the new system incrementally; and it provides fallback capability, as actions can be reverted to SRP execution if critical issues are discovered during migration.

The primary architectural trade-off of this approach is that the catalog configuration must be maintained in the SRP database during the coexistence period, as SRP lacks the sophisticated configuration features introduced in SRM (three-level hierarchy, external API configurations, template-based autotill). The SRM system operates as a read-only consumer of catalog data from SRP, applying its advanced configuration features as an overlay on the imported base catalog.

4.2 Catalog Synchronization from SRP to SRM

The catalog synchronization mechanism implements a unidirectional data flow where SRP serves as the authoritative source of catalog definitions, with SRM periodically importing and enriching this data with its extended configuration capabilities. This synchronization is executed through the `sync_srp_catalog` Django management command, scheduled as a Kubernetes CronJob for daily execution at 01:00 AM.

4.2.1 Synchronization Architecture

The synchronization process establishes a direct ODBC connection to the SRP SQL Server database and executes a series of import operations that reconstruct the complete catalog hierarchy in the SRM PostgreSQL database. The process imports action types, categories, groups, customers, actions, configuration parameters, fields, and customer-specific customizations in a carefully ordered sequence that respects foreign key dependencies.

The synchronization executes with audit logging disabled to prevent creation of thousands of audit log entries for bulk import operations, improving performance and reducing database bloat. Each synchronization function implements a consistent pattern: query the source table in SRP, construct Django model instances for new or modified records, and execute bulk create/update operations.

4.2.2 Differential Synchronization Logic

The synchronization functions implement sophisticated differential logic that minimizes database operations by detecting actual changes rather than blindly recreating all data. This approach constructs in-memory dictionaries of existing records keyed by primary key, then iterates through source records to segregate them into create and update lists. Bulk operations execute with batch sizes appropriate for PostgreSQL, significantly outperforming individual save operations.

An initial full synchronization strategy that deleted and recreated all records

at each execution proved inefficient with SRP's large data volumes, resulting in execution times around 2 hours. The current differential logic significantly reduces database load, bringing average execution times down to approximately 20 seconds.

The synchronization includes a deletion phase that identifies records present in SRM but absent from SRP, removing obsolete catalog entries to maintain consistency. This ensures that deactivated actions or removed fields in SRP are properly reflected in SRM.

4.2.3 Relationship Resolution and Foreign Key Management

Catalog synchronization must handle complex foreign key relationships where referenced entities may not yet exist when dependent records are processed. The synchronization order (types → categories → groups → customers → actions → fields) ensures that parent entities exist before children reference them. For many-to-many relationships and optional foreign keys, the sync functions implement validation that skips records referencing non-existent entities, logging warnings for manual investigation.

This pattern prevents foreign key constraint violations while maintaining referential integrity, with nullable foreign keys set to None when references cannot be resolved. The approach balances data integrity requirements with the pragmatic reality of legacy system inconsistencies.

4.2.4 Post-Import Enrichment

Following the base catalog import from SRP, the synchronization process executes several enrichment operations that apply SRM-specific configurations:

API Configuration Synthesis: Creates `ExternalAPIConfiguration` records that define option sources for select fields, mapping SRP's simpler parameter-based options to SRM's sophisticated external API integration framework.

Field Duplication Resolution: Identifies and resolves scenarios where multiple fields with identical names exist within the same action scope, a situation that would cause ambiguity in field references during form rendering and script parameter generation.

HDA Field Configuration: Synthesizes configuration for HDA external fields and classification fields based on ticket type mappings, creating the meta-data required for the frontend to render these additional input controls.

Customer-Specific Customizations: Processes SRP-specific configuration

tables and translates them into SRM's three-level configuration hierarchy, creating `CustomerAction` and `CustomerActionField` records that override default behavior.

The synchronization concludes by resetting PostgreSQL auto-increment sequences to prevent primary key conflicts on subsequent manual record creation, ensuring that future inserts receive non-conflicting IDs.

4.3 Execution History Synchronization

While catalog synchronization flows unidirectionally from SRP to SRM, execution history must be merged from both systems to provide unified visibility into all service requests regardless of which execution pipeline processed them. The `sync_srp_sr_history` command implements this bidirectional synchronization, scheduled to execute at 02:00 AM (one hour after catalog sync to ensure current configurations are available).

4.3.1 Service Request Import Architecture

The execution history synchronization imports three primary entity types: service request headers, field values, and execution logs. The synchronization is filtered by a configurable date threshold to avoid importing the complete historical archive, defaulting to the first day of the current month. This date-based filtering imports only recent execution history while avoiding performance degradation from processing years of historical data. The threshold is configurable through environment variables to support one-time historical imports or adjusted synchronization windows.

4.3.2 Field Value Encryption and Sensitive Data Handling

Service request field values imported from SRP must respect SRM's enhanced security model where sensitive fields (passwords, credentials) are encrypted at rest. The synchronization function implements transparent encryption during import, checking each field's sensitivity flag and applying encryption to values from sensitive fields. This encryption occurs transparently during synchronization, ensuring that legacy SRP data benefits from SRM's enhanced security protections without requiring modifications to the source system.

4.3.3 Log Unification and Type Mapping

Execution logs stored in SRP use a numeric type code system incompatible with SRM's enumerated log types. The synchronization implements a mapping table that translates SRP type codes to SRM equivalents (Creation, Approval, Execution Started, Execution Completed, etc.). This mapping preserves log

semantics across systems, enabling unified filtering and analysis of execution events regardless of their originating system.

4.4 Dual Execution Flow Management

The coexistence architecture's core mechanism is the dynamic routing of service request execution based on action configuration. When a service request is created through the SRM API, the system evaluates the action's `target_type` to determine whether to invoke the new or legacy execution pipeline.

4.4.1 Legacy Flow Invocation

For actions configured with `target_type = SRP`, the service request creation serializer implements specialized handling that creates the HDA ticket but delegates service request entity creation to SRP. This pattern creates the ticket through SRM's HDA integration (ensuring consistent ticket creation regardless of execution flow), but returns a non-persisted `ServiceRequest` object to the API consumer. The actual service request record will be created by SRP's legacy REST API when HDA invokes it with the ticket reference, then synchronized into SRM's database by the nightly execution history synchronization job.

4.4.2 Modern Flow Execution

For actions configured with any non-SRP target type, the service request creation proceeds through the complete SRM pipeline: the `ServiceRequest` record is persisted immediately with `NEW` status, the ticket reference is stored, and the asynchronous Celery task chain is initiated to orchestrate execution through Provision Prime and Ansible. This execution follows the complete workflow described in previous chapters, including validation, state transitions, approval routing, and execution monitoring.

4.4.3 Transparent Frontend Integration

The dual-flow architecture remains transparent to frontend applications, which interact exclusively with SRM APIs regardless of the underlying execution mechanism. The service request creation endpoint accepts identical payloads for both SRP and SRM actions, with the backend handling execution routing internally. This abstraction ensures that frontend applications require no modifications as actions are gradually migrated from SRP to SRM execution.

4.5 Pilot Strategy and Progressive Rollout

The migration strategy implements a risk-managed progression that validates the new architecture incrementally before full deployment, minimizing exposure to potential issues while building operational confidence.

4.5.1 Phase 1: Internal Pilot

The initial migration phase configures a small subset of actions used exclusively by internal IT operations teams to execute through the SRM pipeline. These actions are selected based on low execution volume (limiting impact scope), non-critical business processes (tolerating potential issues), and comprehensive logging requirements (benefiting from SRM's enhanced observability). Internal teams provide direct feedback on execution behavior, error handling, and operational experience, enabling rapid iteration on issues discovered during real-world usage.

4.5.2 Phase 2: Customer Pilot

Following successful internal validation, the migration expands to a carefully selected cohort of pilot customers. Pilot customers are chosen based on technical sophistication (ability to articulate detailed issue reports), diverse action usage patterns (validating various workflow types), and strong relationship with the company (tolerance for minor issues during pilot phase). Each pilot customer has a subset of their actions migrated to SRM execution, typically starting with low-risk operations like information queries before progressing to more critical provisioning workflows.

The pilot phase implements enhanced monitoring including dedicated Grafana dashboards tracking pilot action execution metrics, daily review meetings to assess execution success rates and identify issues, and direct communication channels with pilot customers for immediate feedback. Issues discovered during pilot are addressed through hotfix deployments, with actions reverted to SRP execution if critical problems cannot be resolved quickly.

4.5.3 Phase 3: Progressive Expansion

Successful pilot completion triggers progressive expansion where additional customer cohorts are migrated on a weekly cadence. Each expansion wave includes a defined set of customers and actions, with execution metrics monitored for anomalies before proceeding to the next wave. The expansion velocity is dynamically adjusted based on observed success rates: high success rates (>99% execution completion without errors) enable acceleration, while elevated error rates trigger pause for investigation and remediation.

The rollout maintains a progressive migration of action types from simple

to complex: read-only information queries migrate first, followed by non-destructive provisioning operations (user creation, group membership addition), then critical operations (password resets, account deletions), and finally complex multi-step workflows. This ordering ensures that the most fault-tolerant operations migrate first, building confidence before tackling higher-risk scenarios.

4.5.4 Phase 4: SRP Decommissioning

Migration completion occurs when all customers and actions have successfully transitioned to SRM execution, validated through sustained periods (30+ days) of stable operation without SRP fallback. At this point, the SRP system transitions to read-only mode, maintaining historical data for reference while no longer accepting new service request creation. The catalog synchronization job is deactivated, with SRM becoming the authoritative catalog source. Finally, the SRP infrastructure is decommissioned after a retention period ensuring all audit and compliance requirements are satisfied.

4.6 Data Integrity and Consistency Mechanisms

The coexistence architecture implements several mechanisms to maintain data integrity and consistency across the distributed system during the migration period.

4.6.1 Transactional Boundaries

Synchronization operations execute within database transaction contexts that ensure atomicity: either all records in a synchronization batch commit successfully, or none do, preventing partial updates that would leave the catalog in an inconsistent state. The use of bulk create and update operations with explicit field lists ensures that only intended modifications occur, avoiding inadvertent overwrites of SRM-specific enhancements applied on top of imported SRP data.

4.6.2 Idempotent Synchronization

The synchronization logic is designed for idempotency, enabling safe re-execution in case of failures. Each sync function can be invoked multiple times with identical results, as record identification is based on deterministic primary keys from SRP. Failed synchronization jobs can be retried without concern for duplicate record creation or data corruption, with the differential logic ensuring that only actual changes trigger database operations.

4.6.3 Referential Integrity Validation

Foreign key relationships are validated before record creation to prevent orphaned references. The synchronization order ensures parent entities exist before children, while optional relationships are nulled when references cannot be resolved. Post-synchronization verification queries identify any referential integrity violations that escaped validation, with alerts triggered for manual investigation and resolution.

4.6.4 Audit Trail Preservation

While audit logging is disabled during bulk synchronization for performance, the synchronization jobs themselves are comprehensively logged through the monitoring infrastructure. Execution start/end times, record counts, errors encountered, and performance metrics are recorded in New Relic, providing complete visibility into synchronization behavior and enabling trend analysis to detect degradation or anomalies.

4.7 Operational Considerations and Challenges

The coexistence strategy introduces several operational challenges and architectural trade-offs that require careful management during the migration period.

4.7.1 Dual Maintenance Burden

The primary operational challenge is the requirement to maintain catalog configurations in the legacy SRP system during coexistence. Any configuration changes (new actions, field modifications, customer onboarding) must be performed in SRP and validated through the synchronization pipeline before becoming effective in SRM. This dual-system maintenance increases operational complexity and the potential for configuration drift if synchronization jobs fail or are delayed.

4.7.2 Synchronization Latency

The nightly synchronization schedule introduces up to 24-hour latency between configuration changes in SRP and their visibility in SRM. For urgent configuration updates required to resolve production issues, manual synchronization execution or direct database modification in SRM may be necessary, introducing risk and complexity. The project mitigates this through careful change management processes and pre-planned configuration updates executed during maintenance windows.

4.7.3 Execution History Fragmentation

During the coexistence period, service request execution history is split between SRP and SRM systems, with the unified view available only in SRM after synchronization. Operational staff must be trained to query SRM exclusively for execution history rather than accessing SRP directly, ensuring consistent data interpretation. The synchronization job's recency filter means that very old historical data may not be available in SRM, requiring retained access to SRP for historical research.

4.7.4 Performance Considerations

The synchronization jobs execute significant database operations that consume computational resources and database connections. The jobs are scheduled during low-traffic periods (early morning) to minimize impact on production operations, with timeout configurations adjusted to accommodate long-running synchronization of large datasets. Monitoring ensures that synchronization completion times remain within acceptable bounds (< 30 minutes), with alerts triggered if execution duration increases beyond thresholds indicating performance degradation.

4.8 Migration Benefits and Risk Mitigation

The coexistence and gradual migration strategy provides a pragmatic path to modernizing the service request management infrastructure while maintaining operational continuity and minimizing risk.

4.8.1 Risk Reduction Through Incremental Validation

The dual-flow architecture enables incremental validation of the new system in production environments, building confidence through measured progression rather than high-risk wholesale replacement. Issues discovered during pilot phases affect only limited subsets of customers and actions, with rapid fallback to legacy execution available if critical problems emerge. This progressive validation approach significantly reduces the risk of system-wide failures that would impact all customers simultaneously.

4.8.2 Knowledge Transfer and Operational Readiness

The phased rollout provides operations teams with extended periods to develop familiarity with the new system architecture, monitoring tools, and

troubleshooting procedures. Internal pilot phases enable IT staff to gain hands-on experience before customer-facing deployments, while customer pilot phases validate operational procedures under real-world conditions. This progressive knowledge transfer ensures operational readiness before full-scale deployment.

4.8.3 Business Continuity Assurance

The coexistence strategy eliminates migration-related service disruptions by maintaining the legacy system as a fully functional fallback throughout the transition period. Critical business processes continue executing through proven legacy workflows while lower-risk operations migrate to the new platform. This approach ensures that service level agreements are maintained throughout the migration, avoiding the business impact of failed "big bang" migrations.

4.8.4 Architectural Validation

The gradual migration provides real-world validation of architectural decisions under production loads and conditions. Performance characteristics, scalability limits, integration points, and error handling mechanisms are validated incrementally, enabling refinement based on operational feedback before full deployment. This iterative validation reduces the risk of architectural flaws only becoming apparent after complete migration, when remediation would be significantly more costly and disruptive.

While the strategy introduces operational complexity through the need for synchronization and dual-system maintenance, these costs are justified by the substantial risk reduction and business continuity assurance provided by the gradual approach. The coexistence period represents a necessary investment in migration safety that protects both the company and its customers from the substantial risks inherent in large-scale system replacements.

Chapter 5

Recommendation system through LLM and Retrieval Augmented Generation

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5.4.2 Embedding Strategy

Conclusions

- Reflections on the internship experience and acquired skills
- Considerations on the future of the project and potential developments

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